

1 Rule of Thumb: Explaining Artificial Intelligence 2 Systems using Partial Information

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6 ABSTRACT

7 Explainable artificial intelligence (XAI) seeks to explain how an AI system arrived at a particular output. Many common XAI methods function by measuring the sensitivity of an AI output to changes in various model inputs, requiring unfettered model access to make multiple model inferences per explanation. This is computationally expensive, vulnerable to adversarial attacks, and unviable if a third-party restricts model access for an AI audit. New approaches are hence necessary to enable XAI in emerging applications: (a) zero shot classification using large language models (LLMs), (b) AI auditing of opaque black-box systems that do not provide model access, and (c) the use of AI in scientific discovery. We propose “Rule-of-Thumb” (RoT) explanations – a new approach to XAI based upon a novel mathematical formulation that identifies the most relevant features for predicting the behaviour of a machine learning (ML) model, for a particular datapoint. We show how RoT is well suited to enable XAI in each emerging application listed above. Additionally, RoT meets specific requirements from leading AI regulations, provides a familiar interface and visualisations for XAI practitioners, is completely model-agnostic and makes no causal assumptions, and is significantly faster in practice than alternative XAI algorithms. Link to our code: <https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Artificial-Intelligence-Systems-using-Partial-Information-3E03>

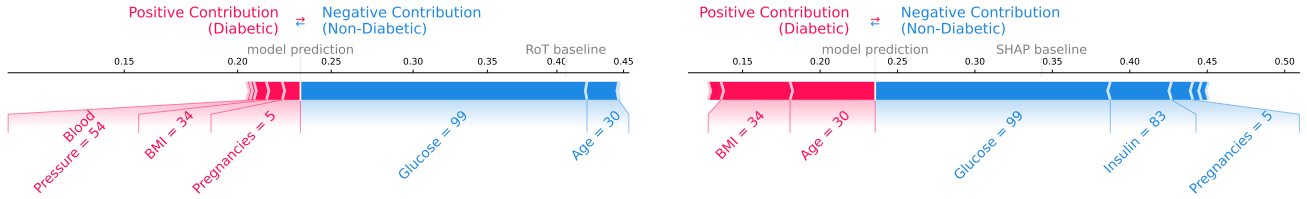
8 To understand how explainable AI (XAI) could work, we consider an example from one of the earliest texts on medical
9 statistics³⁵. Suppose we wished to know where a doctor was on Wednesday night. If we know that most Wednesday nights,
10 they visit the cinema, we might assume they are there. However, if we know they are currently housebound with a broken
11 leg, we would become confident that they were at home. Meehl³⁵ argued that one advantage of an expert over the statistical
12 models of the time, was that an expert could take advantage of these rare events to change their predictions. Inspired by this, we
13 propose RoT, a new form of XAI that quantifies the predictive importance of each feature value, or piece of information, about
14 a particular datapoint.

15 This is a clear departure from existing approaches. Popular approaches to eXplainable AI (XAI) are variants of sensitivity
16 analysis. The underlying intuition behind these approaches is that by perturbing the inputs to a machine learning model, we can
17 identify those inputs that the model is most sensitive to, i.e. the input which upon modification, causes larger changes in the
18 model output. These explainers such as LIME⁴⁴, SHAP³³, and many more^{47,48,50,54} provide post-hoc local feature-importance
19 explanations which describe how sensitive a model output is to each input feature for a particular datapoint. Explanations
20 generated using these explainers are signed vectors representing the importance of each input to the output for a given prediction.
21 Benchmarks to evaluate explainers also often work by verifying that model sensitivity or model gradients match purported
22 feature importance scores. These forms of explanation are increasingly being used to debug models during development, audit
23 black-box AI systems after deployment, or even to understand model training data to propose scientific hypotheses.

24 Such approaches have received substantial criticism. As each feature is artificially perturbed, sensitivity analysis relies on
25 the response of the model to synthetically generated data. This data generation fails to capture causal dependencies between
26 variables, and can create out-of-distribution data that could never occur in the real world. Performing sensitivity analysis
27 implicitly relies on at least three fundamental assumptions: 1. it is possible to pass new datapoints to the model and see how the
28 output alters; 2. perturbing the input results in meaningful changes of the output; and 3. these changes in model output also
29 occur with real-world data.

30 These assumptions make existing forms of XAI inapplicable in three of the most important use cases. 1. *Auditing external*
31 *models* where we do not have control over the inputs to the model. 2. *Zero-shot classification using large language models*
32 *(LLMs)* where models output boolean yes or no responses based on input text rather than the continuous scores required for
33 sensitivity analysis to work reliably. 3. *Scientific Discovery using XAI* where AI models are trained on real-world scientific
34 data, and explanations are sought to uncover natural phenomena rather than explain the AI model. Because the models
35 trained may fit the data without matching real-world causal dependencies, their explanations do not faithfully capture scientific
36 relationships. The SHAP explanation in Figure 1 can only propose valid scientific hypotheses to the extent that the trained AI
37 model accurately predicts behaviour on never seen before out-of-distribution data.

38 We consider an alternate approach that bypasses these limitations. Rather than formulating XAI in terms of how changes in



(a) Rule of Thumb (RoT) explanation for a datapoint. Glucose and age (in blue) contribute most to reducing the predicted probability
(b) SHAP explanation for the same datapoint. Glucose (in blue) is still most important in reducing output probability, but here age (in red) increases it

Figure 1. Examples of local feature importance explanations from a diabetes prediction Random Forest model. The model is trained on the Pima dataset⁵¹ and for a datapoint with feature values: glucose level = 99, age = 30, BMI = 34, etc, predicts a diabetes probability of 0.235. RoT and SHAP explain this prediction differently, but glucose is the most important input feature for both. Standard force plots visualise how individual feature importances contribute to the model prediction.

feature value alter model output, we ask how knowledge of particular feature values should alter our predictions of the model's behaviour. We see echos of this idea in an early work on medical statistics³⁵ that argued that one of the benefits of an expert's prediction over that of a general statistical model is that the expert can exploit important information that occurs too rarely to be captured by a general model. Following this, an unconsidered form of XAI would be to highlight the information most useful for an expert trying to predict model behaviour.

RoT Formulation

We assume that our expert has already uniformly sampled an unknown subset of features¹ with probability p , and wishes to know which features are most likely to be useful to look at next. We make some simplifying assumptions to answer this efficiently. Given a model $C(\cdot)$ defined over features $\mathbb{J}$, and dataset X , we want to find the optimal set of additive functions such that for any subset of features $J \subseteq \mathbb{J}$ and datapoint $x \in X$ we can estimate $C(x)$ as

$$C(x) \approx F \left(\sum_{j \in J} f_{\theta_j}(x_j) + G \right) \quad \forall J \in \mathcal{P}(\mathbb{J}) \quad (1)$$

where $f_{\theta_j}(x_j)$ is a learnt function representing the information provided by knowing that the j^{th} feature takes value x_j , and G a global bias term that indicates what prediction should be made without knowing anything about the datapoint x . We refer to $f_{\theta_j}(x_j)$ as the importance of a feature j taking value x_j . This corresponds to how much a prediction of $C(x)$ should update, on average, after discovering this information.

Given any random subset of features, we can always approximate $C(x)$ using the sum of importances. This idea of always having a rough estimate given any amount of limited information gives rise to the name of the method *Rule of Thumb*. To find the optimal weights $\theta_j \forall j \in \mathbb{J}$ for our model, we minimize the following objective $\mathcal{L}$.

$$\min_{\theta} \mathcal{L} = \min_{\theta} \sum_{J \subseteq \mathbb{J}} \sum_{x \in X} \left(p^{|J|} (1-p)^{(|\mathbb{J}|-|J|)} \cdot \ell \left[C(x), F \left(\sum_{j \in J} f_{\theta_j}(x_j) + G \right) \right] \right) \quad (2)$$

Here the term $\ell \left[C(x), F \left(\sum_{j \in J} f_{\theta_j}(x_j) + G \right) \right]$ is a standard training loss such as log loss or squared error that penalizes mispredictions of $C(x)$. This loss is not only marginalized over all datapoints, as is common in standard machine learning, but also all possible masks of the data. Although computing $\mathcal{L}$ exactly is expensive and requires computing the standard loss over the entire dataset $2^{|\mathbb{J}|}$ times, it can be efficiently optimized using stochastic gradient descent with dropout⁵² applied to the feature importances $f_{\theta_j}(x_j)$. Using dropout, the training time for fitting is almost the same as fitting the simple additive form of equation (1) that does not consider subsets of features.

After fitting the model, the feature j with the largest absolute value $|f_{\theta_j}(x_j)|$ is the one that will most change our confidence in the prediction, and by sorting the features by absolute value, we obtain a

Comparison with existing methods Gradient-based methods^{47,48,50} are computationally efficient but require direct access to model weights, which are often proprietary. With LLMs it is common to provide access only via an application programming interface (API) access. Gradient based approaches cannot be applied to any model that only offers API access.

Sensitivity based approaches LIME and SHAP optimise a per-datapoint objective. While RoT masks information and fits one model for the entire dataset, these approaches perturb datapoints and fit one linear model per datapoint. This creates

¹Using a known feature subset would require each user to specify which features are known, making the explanation vary from user to user.

substantial differences in applicability – not only is sensitivity analysis badly suited for analyzing opaque models that only return yes/no decisions (such as those provided by an LLM API), but it also requires many times (typically $\times 500$ to $\times 5000$) more model or API calls to explain each prediction. Finally, by definition, sensitivity analysis can explain models only in terms of their inputs. RoT can utilise non-input features by fitting on extended datasets if required. This enables explaining model predictions using external models such as BERT⁹ to generate more informative explanations. See **METHODS** for details.

Of existing approaches, RoT is closest to SHAP. While RoT reveals how informative it is to know on average, that a feature takes a certain value, SHAP reveals the average change that occurs upon altering a feature to a certain value from its mean. This is valuable because it allows us to visualise RoT explanations using the same familiar plots as SHAP explanations (eg. the force-plots in Figure 1). This relationship also makes SHAP a natural basis for experimental comparisons with RoT.

XAI for Regulation and Auditing

Compared to other XAI methods, RoT has a range of strengths that make it an ideal tool for making existing AI systems auditable and accountable. Its robustness against adversarial attacks and fidelity compared to sensitivity-based approaches like SHAP give it a clear advantage in meeting the EU AI Act’s requirements for accuracy, robustness, and adversarial testing of high-risk AI systems under Article 15¹⁹. Its robustness against adversarial attacks and fidelity compared to sensitivity-based approaches like SHAP give it a clear advantage in meeting the AI Act’s technical requirements for robustness and resilience under Article 15 (4) – (5). This in turn, supports transparency and instructions of use requirements under Article 13 (1) and (3) (b) (ii). RoT’s computational efficiency compared to approaches requiring construction of a mimic model give it clear advantages in meeting environmental sustainability goals²⁶. Most importantly, it allows auditing proprietary models, overcoming inherent transparency limitations found in many AI regulatory frameworks⁶⁰.

Enforcement of existing AI regulation often falters due to a lack of evidence, stemming from weak or missing requirements for transparency and model access for regulators, third-party auditors, and researchers^{27,37,61}. Gathering sufficient evidence of potentially illegal model behaviour is highly difficult without full API access that would allow regulators to feed the model data to evaluate its responses or oversee model outputs holistically^{6,59,62}. By working without model call access and requiring only observations of prior model outputs or predictions, RoT can bypass significant barriers to regulatory enforceability created by trade secrecy and confidentiality. RoT thus democratizes access to explaining and auditing AI systems. This is crucial because strong provisions enabling access or auditing by third parties (e.g., civil society organizations, vetted researchers) are uncommon in AI regulation, with Article 40 of the EU Digital Services Act being a rare exception^{18,60}. RoT enables participatory governance of both narrow and general-purpose AI systems, facilitating “independent external testing” and adversarial testing of general-purpose systems with systemic risk, as required under Recital 114; Article 55 (1) (a) of the AI Act¹⁵.

In this regard, RoT may prove particularly valuable for enforcement of the AI Act. While Article 74(13) grants market surveillance authorities access to the source code of a high-risk AI system under a limited set of circumstances², Article 74(12) grants market surveillance authorities access to the training, validation, and testing datasets to carry out investigations into these systems. Importantly, this is limited to dataset access, and normally does not include full model access or enable regulators to feed new data to a model to check its responses³. RoT offers a clear advantage over other methods in this regard because it can compute accurate explanations using only these datasets, while others require full API access or training a mimic model.

RoT can also be a powerful tool to identify and challenge anti-competitive practices of major technology companies that are increasingly embedding AI systems across their platforms. RoT can help identify self-preferencing practices, such as search result manipulation, that may be illegal under horizontal competition laws and the EU Digital Markets Act (DMA) that require establishing a link between model behaviour and anti-competitive practices^{16,24}. Establishing causality is, however, not universally required by competition law. A recent case against Google in the United States has shown, for example, that proving the mere existence of anti-competitive conditions is sufficient to show the law has been breached⁵⁶. In this context, even if specific instances of this behaviour in a production environment (e.g., actual transactions with customers) cannot be shown due to a lack of access to model outputs, RoT can demonstrate that a model displays anti-competitive behaviour. RoT can help demonstrate that model behaviour is likely to contravene the competition law prohibitions, and thus assist in the enforcement of competition law. Unlike measuring the sensitivity of mimic models, RoT makes no causal assumptions and thus its ability to predict behaviour of models directly makes it a stronger tool to demonstrate potentially discriminatory behaviour.

²One such circumstance being that the testing, auditing and verification procedures and the provider’s documentation have been “proved insufficient”

³It is worth noting that full API access may be granted under the AI Act to general-purpose AI models (e.g., LLMs) if specific conditions are met and a request made by the European Commission. Specifically, Article 92 (1), (3) – (4) grants the EU AI Office the power to conduct evaluations of the general-purpose model if the EU Commission requests API access (or source code). However, even in such cases RoT has clear advantages because it is one of the only computationally viable methods shown to reliably explain zero-shot classification tasks in LLMs at a meaningful scale.

117 Rule-of-Thumb: Explanations designed for Emerging XAI Applications

118 Explanations obtained through sensitivity-based XAI techniques often serve as an aid for users and developers in understanding
119 and debugging model behaviour. However, they fall short in key emerging application areas: zero shot classification with
120 LLMs, auditing black-box AI, and scientific discovery using AI.

121 Model Evaluation and Zero-Shot Classification with LLMs

122 LLMs are often used for zero-shot classification when there is no labelled dataset to train a classifier model. For each datapoint,
123 an LLM is prompted to output a classification. Since there are no target labels to compare predictions with, typical model
124 performance metrics such as accuracy, precision or recall cannot be readily computed. XAI offers a complementary path to
125 performance evaluation, allowing us to inspect and verify that the “correct” features are predictive of the model’s behaviour.

126 For example, fine-tuned language models can be used to predict the outcomes of judicial appeals from a court summary⁴⁰,
127 or commercial LLM APIs can be used to filter resumes⁵⁸. We consider these two applications, where a key challenge is that
128 new predictions can be difficult to obtain for arbitrarily modified model inputs. When using commercial models, this manifests
129 as increased costs (by default, SHAP requires 500 additional API calls to explain a single API prediction, while LIME requires
130 5000). Even with an open source model, this causes explanations to be prohibitively slow to compute. By contrast RoT only
131 requires a single API call for each datapoint. To improve RoT’s predictive power on text, we generate embeddings for each
132 input token and treat these as the features, performing dropout at the token level.

133 **Explaining Judicial Case Predictions from a Fine-Tuned RoBERTa Model** To test RoT explanations on LLMs, we study
134 an AI system to predict appeal outcomes in Indian courts. We present example explanations for a case in Figure 2. The data
135 includes annotation labels for individual segments in each case. These indicate which portions of the text were important to the
136 judgement⁴⁰. This allows us to measure explanation quality by computing the area under a precision-recall curve (PR-AUC),
137 giving a measure of agreement between the importances from XAI methods and the annotations.

Kapur, J. I. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is ... in support of the contention several cases were relied upon,

6. In our opinion on the findings as given the payment in dispute does not ... it did not affect the quantum of salary of the recipient.

The two cases strongly relied upon by the appellant Company were J. P. Hancock v. General Reversionary and Investment Company Ltd. (1919) ... that was the ground for holding the amount to be a proper deduction.

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(a) A datapoint with text segments highlighted as per RoT importances

(b) The same datapoint with segments highlighted as per SHAP importances

Figure 2. A judicial appeal is rejected by a fine-tuned RoBERTa large language model⁴⁰ making zero-shot predictions. This prediction is explained by RoT and SHAP, with more important text segments highlighted with a deeper colour. Human annotation of the segments is indicated in bold, and aligns better with RoT than with SHAP. We present abridged texts here, with unabridged results from RoT, SHAP, LIME, and Integrated Gradients presented in Figures 9, 10, 11, 12, and 13.

138 As a fine-tuned RoBERTa classifier is used here, standard XAI methods such as SHAP and LIME can also be used
139 to explain the existing classifier. Table 1 presents ROC-AUC scores measuring explanation quality across cases. For
140 LIME we reduced the number of background samples from 5000 to 500 because it is prohibitively slow to compute
141 (on Google Colab’s free tier, it would take over 2 years to run LIME with 5000 background samples per explanation
142 over the entire dataset). Details about preparation, explanation generation, and weighted ROC-AUC evaluation can be
143 found in the **METHODS** section, and all results can be analysed via: <https://anonymous.4open.science/w/Rule-of-Thumb-Explaining-Artificial-Intelligence-Systems-using-Partial-Information-3E03/JudicialCaseOutcomePrediction/Code/results.html>
144
145

	RoT	SHAP (500 samples)	Integrated Gradients	LIME (500 samples)	Random
Average ROC-AUC	0.76668	0.74226	0.64082	0.58910	0.46710

Table 1. For legal case outcomes predicted by an LLM⁴⁰, binary annotations indicate which text segments from the judgement are important. Token importance explanations from an explainer can be considered to “predict” these annotations, which we evaluate using the area under the receiver-operating-characteristic curve, weighted by the length per evaluated text segment.

Explaining Movie Review Classifications from an LLM API We conduct a similar experiment with movie reviews, where we explain zero shot sentiment classification behaviour from OpenAI’s GPT-4.1-nano model. When compared with ground truth importance annotations^{10,64} for review sentiments, RoT achieves a weighted AUC-ROC of 0.72, while a random baseline achieves 0.50. Gradient based explainers (Integrated Gradients) cannot be compared, as access to GPT weights is restricted, and perturbation based explainers (LIME, SHAP) are infeasible to apply even at this modest scale. This showcases a unique RoT ability, where we can explain the behaviour of the model both without access to it’s weights and without making additional API calls, simply by looking at predictions already made by the AI. Additional details are in the **METHODS** section.

Explaining Resume Filtering from Proprietary LLMs We examine a resume filtering system built using OpenAI’s commercial GPT-4.1-nano API⁵⁸. This performs zero-shot classification to identify candidates for information-technology (IT) jobs. We use RoT to generate explanations identifying which parts of the text were important to selecting candidates as suitable for an IT position. We augment our text-based features with three attributes – race, gender, and political orientation. This utilises RoT’s unique ability to produce explanations using non-input features. We verify findings from the original study indicating that GPT ascribes negligible importance to race, gender, and political orientation⁴ when making hiring decisions. We evaluate the system with two types of resumes: teachers and IT-workers. The wordcloud in Figure 3 presents the text identified as predictive of selection and rejection across the resume corpus. Additional data and implementation details can be found in the **METHODS** section.

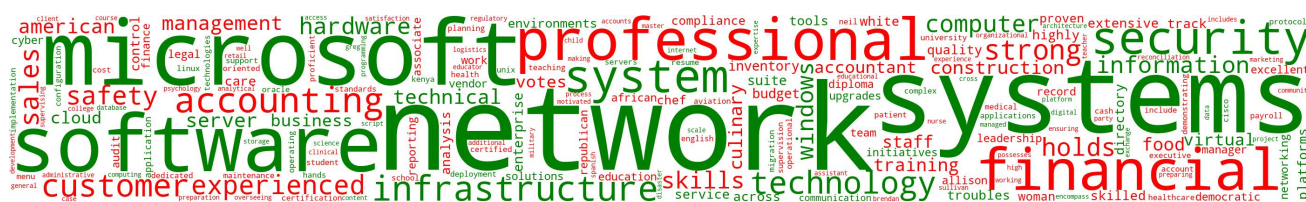


Figure 3. A zero-shot resume filter system uses LLM APIs to find suitable candidates for an information technology (IT) role. RoT explanations indicate the importance of individual text tokens, which we visualise using an importance-weighted word cloud. Alongside IT workers which pass the filter, our corpus contains resumes of bankers, accountants, teachers, and others which our filter rejects. Tokens with positive contributions are in green, and with negative contributions are in red.

Auditing Proprietary Black-Box AI Systems

Explaining *how* a model makes predictions forms a vital component of AI auditing. Explanations reveal model behaviour and are critical for evaluating and auditing AI holistically. When explaining a proprietary AI system, model access can be restricted. With unavailable model weights and the inability to obtain model outputs for synthetically perturbed inputs, no sensitivity-based explainer can be used. One strategy is to collect observable input-output pairs from real-world model deployments, and use this dataset to build a “mimic” AI model. With unrestricted access to the mimic, predictions can be obtained on new synthetically-perturbed inputs, enabling the use of sensitivity based explainers.

In real world settings explaining mimic models instead of the original might be a simple matter of convenience^{2, 20, 22, 36}. Mimic models suffer from the “Rashomon effect” where they can make identical predictions to the original model, but through different internal mechanisms^{42, 43, 46}, consequently yielding different model explanations than the original^{34, 38}. RoT explanations are different, and can be computed without access to the model or the ability to obtain outputs for new inputs, using only the mimic training dataset of input-output pairs. This makes RoT valuable for enforcing regulations which do not grant third-party model or source code access, for example under the AI Act which grants regulators access only to observe model outputs on fixed input data, but not the ability to obtain new prediction on arbitrary inputs by Article 74(12)¹⁹. Similarly, in regulatory contexts where establishing causal links is not necessary, RoT explanations can be used to identify self-preferencing practices¹⁷.

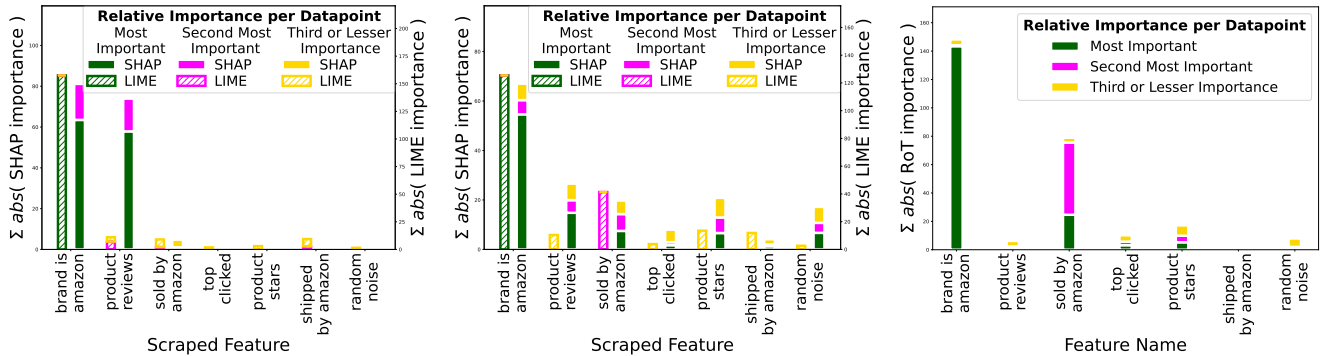
	RF (Markup)	RF (Default)	LogReg	LogReg (L1)	LogReg (L2)
Test Accuracy	0.68551	0.68905	0.71025	0.72438	0.69258

Table 2. Amazon’s product recommendation model is not publicly accessible, making audits difficult. Different models are trained to mimic Amazon predictions, enabling audits by providing query-able interfaces for explainers. We train Random Forest models with hyperparameters from The Markup’s audit or from scikit-learn defaults, and various Logistic Regression models. They all have similar accuracies, forming a Rashomon set of models that make similar predictions.

⁴Following⁵⁸, this is evaluated on synthetic data, by varying names and political orientation on otherwise static resumes.

178 To demonstrate this advantage, we replicate an audit of Amazon’s e-commerce recommendation system conducted by The
 179 Markup³⁰. Lacking access to Amazon’s recommendation system, we scrape product rankings from Amazon and use them to
 180 train a model to mimic how pairs of product are ranked relative to one another. We then explain mimic model predictions. See
 181 **METHODS** for details.

182 Standard explanations reveal the most sensitive input feature is whether the product brand is Amazon (“brand is amazon”),
 183 followed by “product reviews”. However, different mimic models alter feature importances. We find many models in the
 184 Rashomon set of roughly equally performant models, presented in Table 2. We present global (model-level) explanations for
 185 various mimic models from the Rashomon set to verify differences in their internal decision mechanisms in Table 6. Together,
 186 these tables show that models with similar accuracies can have different internal prediction mechanisms.



(a) Explanations from The Markup’s Random Forest (b) Explanations from the scikit-learn default (c) ROT importances obtained without mimic
 mimic model. Using SHAP, they report “brand is Random Forest mimic model differ, but continue to models: “brand is amazon” and “sold by amazon”
 amazon” and “product reviews” to be important indicate the importance of “brand is amazon” are important, “product reviews” is unimportant

Figure 4. Auditing Amazon’s proprietary recommendation model through mimic models (Table 2), and directly with ROT . We visualize explanations via their absolute feature importances, summed over all datapoints. We also indicate the proportion of the importance sum where the feature was the first (in green) or second (in pink) most important for each datapoint. With SHAP, Markup’s mimic model shows brand and product reviews as important features (a). But the default Random Forest mimic model shows different feature importances (b), demonstrating that sensitivity based explainers produce different explanations for different mimics. This undermines the veracity of the audit. By contrast, ROT (c) does not need any mimic models or access to the proprietary model. Extended results are in Figures 15, 16, 17 and 18.

187 In Figure 4 we show how SHAP explanations vary for different Random Forest models, and compare these with the ROT
 188 importances generated without any mimic model. This variation is not due to explainer errors, but rather to intrinsic differences
 189 in model behaviour. An audit that arbitrarily presents results from one mimic model can therefore be undermined by simply
 190 selecting a different mimic. Further, the audit results can be questioned because insights from any explanation can only be
 191 attributed to the chosen mimic, but not the original model.

192 Like previous explanations, ROT also covers “brand is amazon” to be the most important feature. However, critically,
 193 it also shows “sold by amazon” is an important feature, something that remained hidden in The Markup’s original analysis.
 194 This is a new insight signalling a preference for products sold by Amazon over those from third party sellers. Because the
 195 explanation is independent of idiosyncrasies arising from the choice of mimic model, and ROT makes no causal claims, this
 196 insight into potentially self-preferencing behaviour cannot be undermined like others.

197 Scientific Discovery using XAI

198 Model explanations are sought not just to explain the behaviour of a model, but to also understand the underlying data it was
 199 trained on. This strategy has been advocated as a means of scientific discovery^{11, 12, 14, 23, 45} in diverse fields like medicine¹³,
 200 engineering²⁸, chemistry²¹, and many more^{3–5, 8, 25, 41, 53, 63}. Upon analyzing 1151 research articles at the intersection of XAI
 201 and science, we found 4.9% of them used model explanations as a means of novel scientific discovery. Of these, 70% used
 202 SHAP. We detail the methodology behind these statistics in the **METHODS** section.

203 XAI use in scientific discovery suffers the same challenges as AI auditing – like mimic models, model trained to predict
 204 scientific phenomena are not guaranteed to faithfully replicate underlying causal mechanisms. Such uses are criticised because
 205 sensitivity analysis generates explanations by making predictions on synthetic, out-of-distribution datapoints, which may
 206 include physically implausible values. For these implausible, off-manifold datapoints, the behaviour of the model could be
 207 anything, and consequentially the feature importances could be arbitrary^{29, 32, 55}. By contrast, ROT importances are inspired by
 208 feature “predictiveness” rather than sensitivity, overcoming these limitations without implicitly introducing harder problems
 209 such as learning causal models³⁹ or density estimation⁵⁷.

Identifying Uninformative Features as Unimportant If a model shows any changes in outputs for the arbitrary, out-of-distribution perturbations of an input feature, sensitivity based explainers consider that input important³². This definition considers a feature to be important not if it is useful to a model’s predictions, but merely if it causes changes in them, even if the changes themselves are random. In this way, uninformative features can inadvertently seem important. We consider a scenario using the Pima Indians⁵¹ diabetes prediction dataset⁵¹ from Figure 1, transformed using FairPCA³¹ to ensure that the age input feature is statistically uncorrelated with the target output – diabetes.

We consider a scenario where researchers are interested in understanding diabetes and its diagnosis. Explaining a diabetes prediction model is a common way to achieve this⁷. On our transformed dataset, since age has been deliberately rendered uninformative, a useful XAI technique would correctly identify that age does not affect diabetes diagnoses. We use this as a test to determine if an explainer is suitable for use in scientific discovery. Figure 5 displays feature importances from RoT, SHAP, and LIME on transformed diabetes data. RoT is the only explainer to consider age unimportant. Model details and our use of FairPCA³¹ to ensure statistical independence between age and diabetes are detailed in the **METHODS** section.

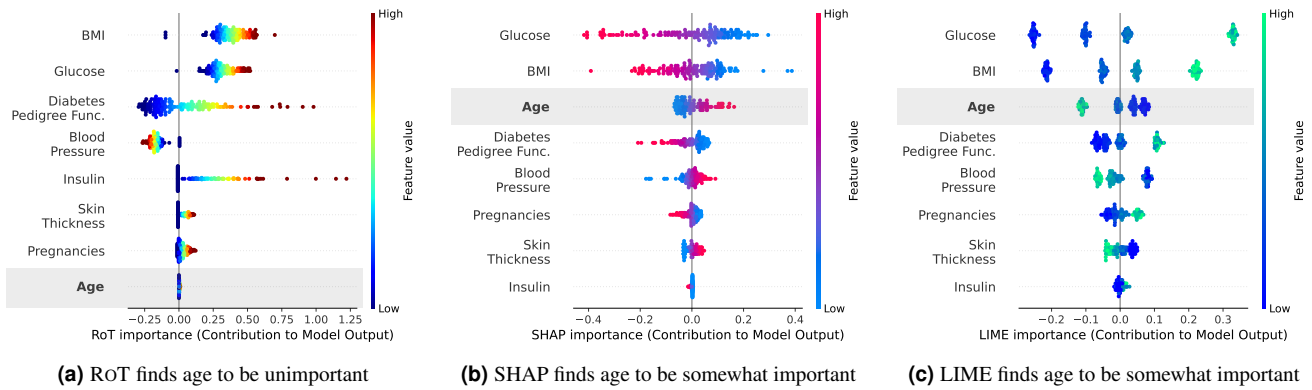


Figure 5. When using XAI for scientific discovery, explanations can help improve our understanding. A diabetes prediction model is trained on synthetically transformed “Pima Indians data”⁵¹ such that age is uncorrelated with diabetes. Explanations should ideally surface that age is unimportant. Swarmplots display feature importances across the dataset. While RoT correctly ascribes negligible importance to age due to our data transformation, sensitivity based explainers SHAP and LIME do find it to be somewhat important. When using XAI for scientific discovery, RoT thus provides clear benefits over SHAP and LIME.

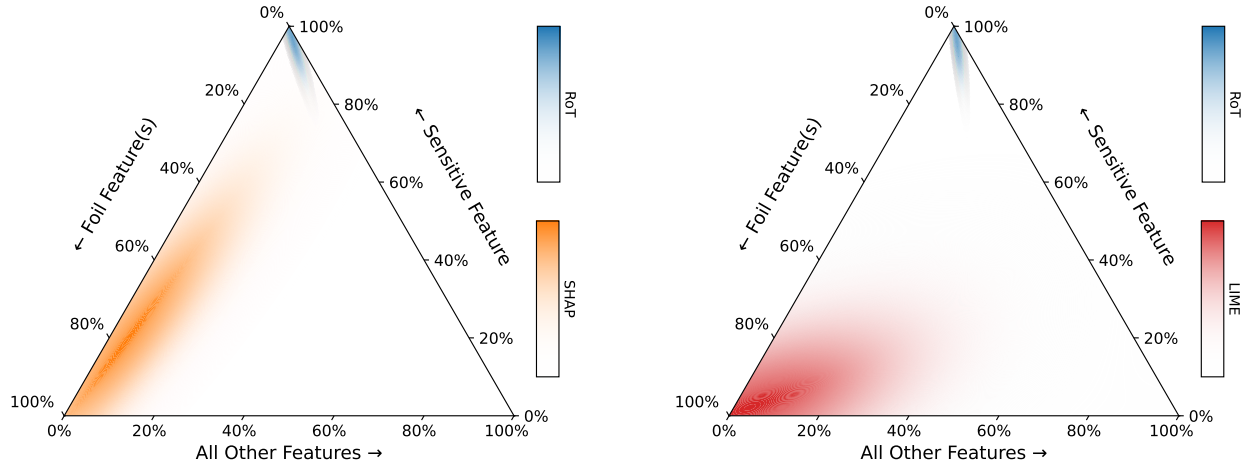
Although age is, by definition, uninformative and unresponsive of the output of the model on the transformed dataset, altering its value in isolation does alter the classifier response, and consequentially SHAP and LIME find it to be an important feature. These explainers are not erroneous – they are just measuring model sensitivity, which does not serve as a good measure of importance for scientific discovery. If our interest is in learning about the laws and relations that govern the data, RoT is the better method. In addition to force-plots from Figure 1, the swarmplots in Figure 5 demonstrate RoT can be used as a drop-in replacement in existing XAI code.

Identifying Important Features in the presence of Misleading Features We have seen that explainers can be vulnerable to out-of-distribution model behaviour. Adversarial attacks have been proposed to exploit this vulnerability⁴⁹ which modify models to explicitly use different internal mechanisms on out-of-distribution data. This effectively presents different model behaviour to SHAP or LIME than to real-world inputs. The adversarial attack masks model behaviour by hiding the actual importances of “sensitive” features and manipulating explainers into selecting misleading “foil” features as important instead.

When using XAI for scientific discovery, the inadvertent inclusion of adversarial models in the Rashomon set can cause misleading results. In black-box AI audits, this can enable “fairwashing” by allowing malicious actors to deliberately hide models with discriminatory behaviour behind benign-looking explanations. To test RoT in such a situation we construct adversarial models designed to hide the importance of an important “sensitive” feature by introducing one or more misleading “foil” features. We then generate explanations and check the relative importance of the sensitive feature compared to the foil features and other features. In Figure 6 we show the distribution of importances over three feature categories: sensitive, adversarial, and other, from an adversarial attack we replicate for SHAP and LIME⁴⁹. Details about the datasets used, model behaviours introduced, “sensitive” input features masked and adversarially misleading “foil” features introduced are available in the **METHODS** section. We report detailed statistics about the proportion of explanations that identified the sensitive feature as most important (success), or the foil feature (adversarial failure) or other features (non-adversarial failure) in Table 7.

Across 10 experiments, for LIME or SHAP the rate of recovering the sensitive feature as most important does not exceed 5% except once, where adversarial failures (i.e., foil feature is most important) still outnumber success (i.e., sensitive feature is

most important). ROT has a success rate always greater than 89%, and in 6 out of 10 experiments, ROT has a perfect recovery rate of 100%, correctly identifying the sensitive feature to be the most important for every prediction.



(a) Distribution of feature importances from SHAP and ROT explanations across “sensitive”, “foil” or other features for an adversarial model

(b) Distribution of feature importances from LIME and ROT explanations across “sensitive”, “foil” or other features for an adversarial model

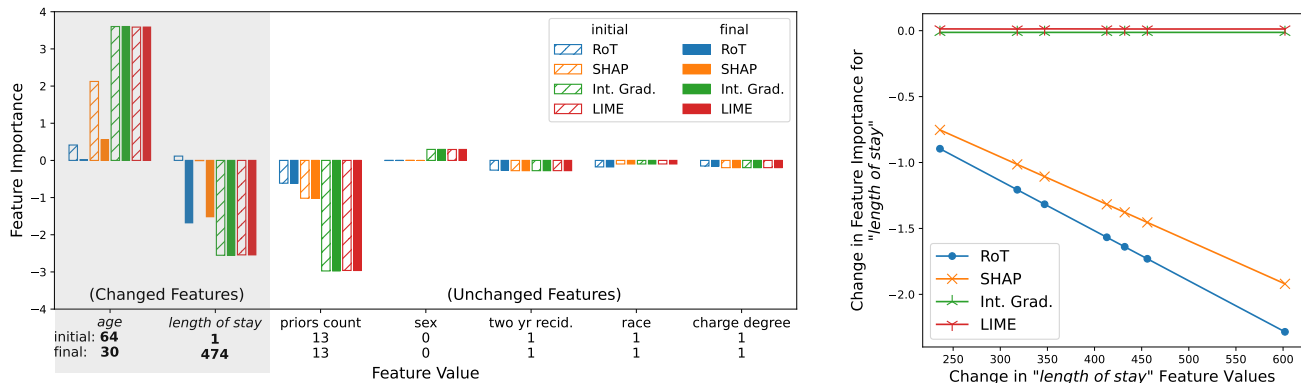
Figure 6. An adversarial attack on explanations where the model makes predictions using a “sensitive” demographic feature, but the adversary misleads explainers into ascribing importance to spurious “foil” features instead. We plot the distribution of feature importances across the “sensitive”, “foil”, and other features, and find that both SHAP and LIME are misled by the adversary, but ROT correctly ascribes importance to the “sensitive” feature.

Benchmark XAI Evaluations and Computational Efficiency

As demonstrated, ROT has clear benefits in emerging XAI applications of scientific discovery, auditing, and LLM explanation. We now analyse ROT in traditional tabular modalities using existing XAI benchmarks, where it is a drop-in replacement for other explainers. We observe ROT produces similar explanations as SHAP but is more computationally efficient.

Benchmark Evaluations of RoT Explanations

Sensitivity analysis underlies XAI evaluation benchmarks just like it underlies explainers. The OpenXAI benchmark¹ consists of explanation evaluation metrics that either perform sensitivity analysis, or comparisons with “ground-truth” explanations.



(a) Feature importances for an initial datapoint and for a final datapoint with a changed prediction. We expect input feature changes to change the corresponding importance, but this happens only for (b) Changes in “length of stay” feature importance for changes in input feature values. Like in (a), ROT and SHAP change but LIME and Gradients are constant

Figure 7. Analysing explainers using a pretrained binary classification model from the OpenXAI benchmark (COMPAS data). For any initial datapoint, if changing a feature value changes the class predicted by the model, that feature should be important. We show all feature importances for one such datapoint (a), and the changes in one importance across all such datapoints (b).

OpenXAI considers Logistic Regression coefficients as the “ground-truth” explanation for all datapoints. Regardless of the specific feature values for a given datapoint, such ground-truth comparison metrics consider a single explanation as ideal across all input datapoints, promoting a single unchanging explanation. Using Logistic Regression coefficients OpenXAI deliberately promotes gradient methods or LIME, which are Taylor-series like approximations of model behaviour and can recover the Logistic Regression coefficients exactly for any input datapoint. While these explainers score better than SHAP in the published OpenXAI benchmark scores, they are undesirable in practice because they cannot explain individual predictions and always produce identical feature importance outputs. Due to these aforementioned issues, instead of reporting the official metrics from OpenXAI, we adopt the benchmark to verify that RoT behaviour mirrors SHAP, and every input datapoint does not yield the same explanation output.

Figure 7 shows how changing input feature values do not always lead to changes in the corresponding feature importances. Only the SHAP and RoT explainers show desirable behaviour, where the local explanation is truly a function of the datapoint being explained. LIME and gradients show the same feature importance regardless of the feature values, undermining the usefulness of OpenXAI metrics which consider these explanations to be the best. All gradient based methods in OpenXAI generated similar explanations, and here we only show Integrated Gradients for clarity. A detailed description of Figure 7 data is in the **METHODS** section.

Runtimes and Computational Efficiency

Many local explainers function by collecting data about model behaviour and then fitting an interpretable model. Sensitivity based explainers like SHAP and LIME collect data by perturbing inputs and fitting a new interpretable model for each explanation. This requires data collection by making arbitrarily time-consuming calls to the model for each additional explanation. By contrast, RoT fits a single interpretable model on the entire unperturbed dataset, and each explanation is generated using per-feature outputs from this model. Since model inference is quicker than training, this makes computing explanations extremely fast once RoT is fit. It also reduces the environmental footprint of computing explanations by eliminating the need to train new interpretable models or generate extra predictions on perturbed inputs. This latter component can be very computationally expensive in modern AI applications, as we noted in our judicial case prediction experiment, where SHAP required significant resources and LIME was rendered completely unviable at scale.

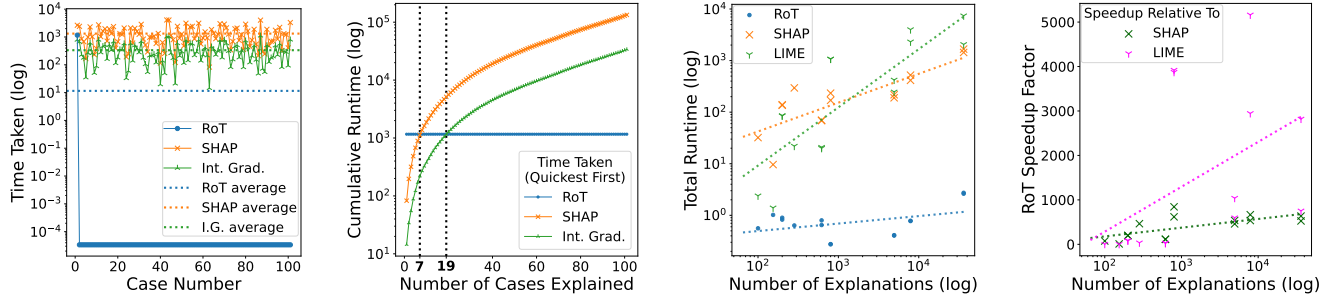


Figure 8. Program runtimes (in seconds) from experiments with text data – (a) and (b); and tabular data – (c) and (d). While the first RoT explanation takes time, additional explanations are virtually instantaneous, unlike SHAP and LIME.

Consequently, the time-complexity of SHAP appears linear in number of cases, while RoT appears to be constant. For the text data, 13 million RoT explanations can be computed for each additional SHAP explanation, after the first. Even the 7 fastest SHAP explanations take as long as all 101 RoT explanations. For the tabular data, the RoT speedup factor increases with explanations as the RoT initialisation gets amortised over more of them, reducing the average time taken. Further, we see this increase faster for LIME than SHAP, reflecting higher marginal costs (5000 background samples each vs 500 for SHAP).

In practice it is impractical to compute SHAP or LIME explanations for LLM predictions, except in rare cases where only a handful of datapoints need explaining. Gradient methods are quicker, but require model access which is also unavailable in practice over commercial APIs. RoT is the first explainer that can actually be deployed in practice, bringing feature importance XAI to LLMs. In Figure 8 we analyse explanation runtimes across experiments, treating text data separately. For judicial case outcome prediction, the first RoT explanation took 1160 seconds, and each additional explanation took less than 0.1 milliseconds. By contrast, the average time taken for each SHAP explanation was 1320 seconds. On tabular data we find similar behaviour, where RoT is orders of magnitude faster than SHAP and LIME, despite including explainer initialisation times.

Conclusion

This paper presents a new approach to explainable AI based on the predictiveness of feature values, rather than their sensitivity to alteration. This *rule of thumb* approach allows us to approximate any decision making process by summing the importance response over any subset of features values. The most useful feature values for a prediction are naturally those with largest importance, and their predictive power overrides the contribution of others. In this way, RoT is a post-hoc local explanation method inspired by ablation studies, rather than sensitivity analysis.

This approach addresses several long-standing concerns in the community about the validity and practical utility of XAI, and importantly does so without making additional assumptions. It is flexible and useful in a range of emerging XAI applications areas where other forms of XAI simply can not be used – evaluating LLMs and explaining their behaviour in zero-shot classification, auditing proprietary AI systems where we can only observe their predictions and not alter input features, and using XAI to understand real-world phenomena and propose novel scientific hypotheses. As a local, post-hoc explanation method, RoT can be used as a drop in replacement for other explainers in existing XAI codebases, and visualised the same way, providing an exceedingly fast and environmentally friendly method of explaining AI model behaviour.

Competing Interests Statement

We have no competing interests.

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METHODS

1 RoT Implementation Details

Recall our formulation from Equation 2. In all experiments, we treat ℓ as the log loss, and F as the sigmoid function, arriving at an objective similar to logistic regression. We consider three variants of the importance function $f_{\theta_j}(x_j)$.

The linear form

$$f_{\theta_j}(x_j) = a_j(x_j + b_j) \quad (3)$$

A more general additive form consisting of a linear form plus a mixture of Gaussians

$$f_{\theta_j}(x_j) = a_j(x_j + b_j) + \sum_{i < I} S_{i,j} e^{-\frac{(x_j - \mu_{i,j})^2}{\sigma_{i,j}}} \quad (4)$$

While we explored the use of other additive models including polynomial kernels and the double exponentiated form proposed in², the mixture of Gaussians combined with a linear term, showed better stability, and robustness to extreme values.

And finally, for explaining the behaviour of text-based classifiers, such as ChatGPT for zero-shot classification, we use a linear form shared over all tokens

$$f_{\theta_j}(x_j) = a \cdot x_j + b \quad (5)$$

Here x_j is now a vector representing the text embedding taken from a model such as BERT, and a and b are common parameters shared over all token locations $j \in \mathbb{J}$. Based on this, we provide two canonical implementations of RoT. One uses dropout on the features and a simple linear model to learn classifier logits or regression outputs using either cross entropy or mean squared error loss, and the other uses text embeddings, performing dropout on both embedding features and tokens, to explain zero shot classification tasks.

Our experiments were performed on a MacBook with 24 GB memory and an Apple M4 Pro Processor. All the code used for our experiments will be made public. We also plan to make RoT available for the community as an open-source software package. We make use of existing implementations of explainers such as SHAP¹¹ and LIME¹⁴, other open sourced data and models^{8,12,17,18}, and evaluation benchmarks and code^{1,16} previously released by the community.

2 Experiments

2.1 Evaluating Zero-Shot LLM Classifier Models with XAI: Explaining Judicial Case Predictions from a Fine-Tuned RoBERTa Model

From the published PredEx¹² data, we start by removing all annotations that contain hallucinations or text that we could not reliably match in the raw case text. We drop datapoints with missing case text or missing annotations, and datapoints where no text segments were found to be important or where the entire case text was found to be important. The data was annotated by law students who tried to identify text responsible for the judgement. However, the finetuned model making predictions does not always match the ground-truth predictions. Since the explanation annotations are produced from the ground-truth, but our explainers try to explain the model, there is a discrepancy. To resolve this, we drop all cases where the model predictions were incorrect, since we are interested in explaining the model but comparing our outputs with annotations which were produced independent of the model. From the remaining subset, we further drop those cases where highlighted segments were too “short” to be reliable: we enforce the condition that the minimum character length of a highlighted or non-highlighted segment in the text should be at least 75 characters. This threshold is large enough to ensure that common phrases do not get highlighted unnecessarily, and small enough to ensure we still have enough datapoints pass the filter and be included in our experiment.

This finally leaves us with a set of 502 cases, from which we compute SHAP and RoT explanations on a random 20% of cases (101 cases total). The RoT explainer we build uses the remaining 401 cases as background, but results using all 502 cases or only the 101 explanation cases as background are qualitatively similar. For the 401 case corpus, RoT is only fit on the final RoBERTa prediction per case, and not on the per-segment annotations within the case text.

We report the PR-AUC of the explanations with respect to the ground truth human annotations (Table 1). Computing this metric involves comparing explanation importances (which can be negative) with human annotations (which are a binary 0 or 1). Since the annotations indicate whether text is important without direction (appeal rejection or acceptance), we post-process the feature importance by aligning them with the case prediction manually. That is, if the case appeal was rejected, we consider negative values to denote higher importance by flipping the sign of the importance for all segments, and if the case appeal was

accepted, we consider the importance values as is. This way, positive values always to indicate higher importance as defined by the annotators. This post-processing is not specific to RoT and is common to all explainers because it is a peculiarity arising from the PredEx data.

For RoT explanations, we use a straightforward implementation where dropout is used both on input features and on input tokens. For SHAP, we use the canonical partition explainer implementation (which defaults to 500 background samples) available for use with language models⁵. For LIME we use the text explainer with its default bag-of-words tokenizer. By default this requires 5000 background samples, but this turns out to be prohibitively slow to compute on the entire dataset so we reduce this to 500 for PR-AUC calculations. In addition to abridged example in Figure 2, program runtimes are documented in Figure 8. The RoT and SHAP explanations for the unabridged case can be seen in Figures 9 and 10. Additionally, LIME and Integrated Gradients explanations for this particular case (we could not compute 5000 sample LIME explanations for all cases because it was prohibitively slow) can be seen in Figures 12, 13, and 11 respectively. Results from other cases can also be analysed via <https://anonymous.4open.science/w/Rule-of-Thumb-Explaining-Artificial-Intelligence-Systems-using-Partial-Information-3E03/JudicialCaseOutcomePrediction/Code/results.html>.

2.2 Explaining Movie Review Classifications from an LLM API

The ERASER⁴ benchmark provides multiple text datasets with annotations indicating importance of constituent tokens. Most of the tasks however consist of retrieval (from an LLM context window), which can be evaluated using these annotations. The movie review dataset²⁰, however, presents an exception. The task here is simple zero-shot classification and does not involve any explicit retrieval.

We use the GPT-4.1-nano API to obtain predictions for movie reviews (positive or negative), which we then seek to explain using RoT. Our dataset consists of 1600 movie reviews from the ERASER benchmark dataset⁴. To explain these LLM review sentiment classifications we repeat our process from explaining fine-tuned RoBERTa model predictions for judicial cases exactly, producing RoT explanations for each text segment from the movie reviews and measuring how well they align with human annotations. The notable difference between the two setups is that when explaining judicial cases, we reused the embeddings from the finetuned RoBERTa model itself, whereas now we use ModernBERT¹⁹.

Choice of Evaluation Metric For both LLM experiments discussed so far, we evaluate explanation quality using the area under the receiver-operating-characteristic (a plot of the true positive rate against the false positive rate). We choose this metric over the area under a precision-recall curve because the precision-recall curve is sensitive to annotation density¹⁵. That is, the precision recall metric is highly sensitive to what proportion of tokens are deemed important for any given text. Since the annotations are booleans, this is dependent entirely on annotator idiosyncracies – we simply do not have ground-truth importance scores per segment). To illustrate the impact of this, consider the areas under the precision-recall curve reported in the top row of Table 3. If we simply flip all the labels in the ground truth annotation and simultaneously flip the signs of the RoT importances (the equivalent of finding / deleting unimportant segments instead of finding / selecting important segments from a given review), the metrics change as is visible in the bottom row of the table.

	RoT	Random
Average PR-AUC	0.58325	0.25870
(flipped) Average PR-AUC	0.94777	0.91689

Table 3. Area under the Precision-Recall curve: Evaluating explanations by comparing them with ground annotations. We compare ground truth labels with RoT importances to measure the performance in terms of identifying important text segments. While RoT is better than random, PR-AUC is sensitive to annotation density, and flipping the labels affects the metric significantly. For this reason we choose to use the AUROC metric instead.

Further, the metrics reported in this work for evaluating LLMs are all weighted by segment length. This follows from the intuition that given two segments misclassified by an explainer (deemed as important when annotated as unimportant or vice versa), we wish to penalise the error for the longer segment more severely. Judging the importance of a single character or token incorrectly is more “forgivable” than incorrectly ascribing importance to a larger text segment making up say half the input passage.

⁵this has been made part of the official SHAP package: https://shap.readthedocs.io/en/latest/text_examples.html, but not documented through any publications

2.3 Evaluating Zero-Shot LLM Classifier Models with XAI: Explaining Resume Filtering using Proprietary LLM APIs

We replicate a study that found GPT-3 to be largely unbiased in resume filtering tasks¹⁸. Starting with a synthetic corpus of resumes consisting of information-technology workers, accountants, aviation workers, construction workers, chefs, advocates, teachers, and salespeople; we use GPT-4.1-nano to select the IT workers from this corpus. In such a scenario where zero shot classifications are obtained from an LLM API which cannot be queried at will, RoT is still able to explain predictions.

Just like explaining case outcome prediction and movies, we are able to produce token level importance scores using RoT. This time, we use BERT³ embeddings instead of ModernBERT¹⁹, showing how our method is embedding agnostic and can be used with various text embeddings. We additionally augment explanations with non-input features so predictions are explained not just in terms of the resume, but also in terms of the demographic attributes of the candidates. The original study used race, gender, and political orientation, and we explicitly add this information to our explanation text (even though the predictions were made using only the resume summary). This is an additional ability unique to RoT, and is particularly useful for black-box algorithmic auditing. It allows us to generate feature importance explanations using a superset of the model inputs – we include race, gender, and political orientation. Our results match the original study, and find that these features have negligible impact on model predictions.

In addition to the wordcloud in Figure 3, we visualise the importance ascribed to a particular token across resume summaries. A cursory examination of these shows that IT related words are positively correlated with IT workers, whereas words belonging to other professions are negatively correlated. Some of these can be seen visualised in Figure 14.

2.4 Auditing Proprietary Black-Box AI Systems without Mimic Models

Journalists at The Markup audited Amazon’s product recommendation system. Since the model used is proprietary, it cannot be queried at will, which renders all previous explainers inapplicable. To circumvent this, product recommendation data was scraped from the Amazon website, consisting of product pairs – the first and second product recommended for any particular search. A model would be built as a “proxy” to mimic Amazon product rankings, formulated as a binary classifier selecting one of the products from the scraped pair. Feature inputs to this model were also scraped, consisting of attributes like product stars (the difference in number of stars between the pair of models), and whether the picked model brand was Amazon, was sold by Amazon, shipped by Amazon, etc.

The Markup analysed this data by building a Random Forest model and then computing SHAP explanations of this model⁶. Following this methodology, they discovered Amazon to systematically promote products from its own brand over others, as SHAP considered “brand is Amazon” to be the most important feature, and “product reviews” to be second. The model trained and explained was not arbitrary, but chosen after a typical hyperparameter search. When we replicated this audit, we found several models that were (marginally) more accurate, which we list in Table 2. We found the scikit-learn default Random Forest had better performance than the Random Forest with hyperparameters chosen from The Markup’s hyperparameter search. We also found the default Logistic Regression in scikit learn to have better performance, as did both regularised Logistic Regressions (C=0.01).

All our models had similar accuracies and formed a Rashomon set. Their internal mechanisms were distinct, as can be verified from the global explanations summarised in Table 6. This variation is also evident from the SHAP explanations for these models, which led to different conclusions about input feature importances depending on the mimic used. All models found “brand is amazon” to be the most important input feature, but following this, subsequent features had different importances depending on the model selected. For brevity, we present SHAP explanations from just Random Forest models in Figure 4, but include results from all models in Figure 15 and Figure 17.

RoT explanations, by contrast, do not require mimic models. Results obtained from RoT are therefore stronger in an auditing context, because they cannot be attributed to peculiarities in any mimic model, which can otherwise be used to absolve the original inaccessible, proprietary model.

2.5 Literature Review to Determine the Prevalence of XAI for Scientific Discovery

We measured how often explainability is used as a means of scientific discovery to understand the natural phenomena that underlie models by performing a literature review.

To do this, we analysed papers fetched from the Scopus database related to various scientific fields that used some form of interpretable machine learning. We queried the Scopus API using the TITLE-ABS-KEY field with a search string constructed as follows:

```
(  
  "explainable artificial intelligence" OR
```

⁶open-source code from The Markup’s reporting: <https://github.com/the-markup/investigation-amazon-brands>, with both the scraped data and hyperparameter tuning steps

```

597     "SHAP" OR
598     "interpretable machine learning"
599 ) AND (
600     "social science" OR
601     "physics" OR
602     "chemistry" OR
603     "biology" OR
604     "geophysics" OR
605     "astrophysics" OR
606     "biochemistry" OR
607     "medicine" OR
608     "neuroscience" OR
609     "scientific discovery"
610 )

```

611 This yielded 1151 articles, which we filtered using LLMs to analyse the abstracts for mentions of XAI for forming new scientific
612 hypotheses. Using the LLM filter, we reduced our corpus to 111 articles, which we checked manually to verify the use of XAI
613 in each paper. We found that 56 of these papers used XAI, and 39 of the 56 did so using the SHAP explainer. These article lists
614 can be found at:

- 615 • https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Artificial-Intelligence-System-LitReview/a.Corpus_1151.csv
- 616
- 617 • https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Artificial-Intelligence-System-LitReview/b.LLM_filtered_111.csv
- 618
- 619 • https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Artificial-Intelligence-System-LitReview/c.Manual_XAI_56.csv
- 620
- 621 • https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Artificial-Intelligence-System-LitReview/d.Manual_SHAP_39.csv
- 622

623 respectively. Our reported statistic of 4.9% papers using XAI for scientific discovery is thus a lower bound – there might
624 be papers using XAI that were missed by our LLM filter and never included in set of 111 papers we manually inspected. If
625 the LLM false negative rate is assumed to be the same as the false positive (55/111) rate, our final result could be as high as
626 49%. Our objective however is not to perform a thorough literature review or identify what proportion of papers use SHAP
627 to generate new scientific discoveries. Instead, we merely wish to demonstrate that a significant amount of modern scientific
628 discovery relies on sensitivity based explainers to generate hypotheses that seek to causally explain real-world phenomena.

629 2.6 Scientific Discovery using XAI: Identifying Uninformative Features as Unimportant

630 When using XAI for scientific discovery, we want explanations not to consider uninformative input features as important. We
631 deliberately transform⁹ the “age” feature of the “Pima Indians”¹⁷ diabetes prediction dataset to be uncorrelated with diabetes,
632 the target feature, and then compute explanations with various explainers to test which ones still find “age” to be important.

633 To make a feature statistically uncorrelated with the target, we adopt the FairPCA⁹ method to transform our dataset. For
634 our dataset with 8 input features, this transformation works by learning a low dimensional space of 7 features, such that age
635 information is deliberately lost. By transforming our data into this space, any model trained cannot utilise age information.
636 Finally, we learn a Logistic Regression classifier on the data, and use standard SHAP, LIME, and RoT explainers to obtain
637 feature importances. As seen in Figure 5, we find sensitivity based explainers unable to identify the unimportance of age, but
638 RoT does this perfectly.

639 2.7 Scientific Discovery using XAI: Identifying Important Features in the presence of Misleading Features

640 We consider a scenario where an adversary tries to mislead explainers into selecting spurious “foil” features as important,
641 instead of the “sensitive” features the model actually uses to make decisions.

642 We use the same datasets as those used in the paper proposing the adversarial attack¹⁶, and follow its setup exactly: for the
643 COMPAS¹⁰ and Communities & Crime¹³ datasets, the “sensitive” feature is “race” whereas for the German Credit⁵ dataset it is
644 “gender”. All models are binary classifiers formulated as simple if statements depending on the sensitive feature, predicting
645 recidivism risk for the former two datasets, and loan repayment probabilities for the latter.

The “foil” attributes used are synthetically added features for COMPAS and Communities and Crime, whereas for the German Credit⁵ dataset, the foil used is not synthetic but a new feature that is a function of other input features. Further, instead of 3 experiments (one for each dataset), for COMPAS and Communities and Crime we conduct experiments twice – once with a single synthetic foil, and once with 2 synthetic foils. The adversary modifies models to mislead explainers. These modifications work can be tailored to the particular perturbations used by a sensitivity based explainer, and we consider models designed to mislead both SHAP (m_S) and LIME (m_L), comparing both with RoT.

Like in the original paper, this setup leads to 5 experiments with each explainer: 1 German Credit, 2 COMPAS, and 2 Communities and Crime. For each datapoint in each experiment, we consider the absolute importance of the sensitive feature, average absolute importance of the foil features, and average absolute importance of all other features, forming a sample from a ternary distribution. We visualise this distribution of importances in Figure 6 using publicly available software^{6,7}, combined from across each of the 5 experiments so each experiment has the same weight. We also present exact numerical results for detailed perusal in Table 7.

2.8 Benchmark Evaluations of RoT Explanations with OpenXAI

We compute explanations of the pretrained OpenXAI models by running publicly available code from the benchmark¹ as provided⁷. We present results from the COMPAS dataset because it has the fewest features from among OpenXAI datasets, allowing us to present results succinctly. To select an initial point and a final point for plot (a), we find pairs of datapoints with different model predictions such that l_0 distance between them is 2 or less, and l_1 distance between them is as large as possible. For plot (b), we consider all pairs where l_0 distance is exactly 1 while maximising l_1 distances as before, and ensuring the feature changed is always “length of stay”. While we present select results, we are able to repeat this process for all provided pretrained models and datasets: COMPAS, Home Equity Line of Credit (HELOC), Adult Income, and German Credit. These provide the runtime statistics seen in Tables 13 and 14 and Figure 8.

3 Data and Code Availability

All code used in our experiments can be found at <https://anonymous.4open.science/r/Rule-of-Thumb-Explaining-Ar>. The scripts rely on permissively licensed publicly available datasets, and instructions for downloading these, where applicable, are included with the code.

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Supplementary Materials

A Motivation and Related Work

A.1 Explainability in Machine Learning, Feature Attribution Methods, and SHAP

Machine learning models range from inherently interpretable (eg. decision stumps, linear regressions, rule-sets, or single nearest neighbour classifiers) to uninterpretable black boxes (eg. random forest ensembles, deep neural networks, or markov random fields). In order to explain black-box models, methods have been developed to produce post-hoc model explanations (eg. saliency maps, counterfactual explanations, anchors, prototypes, etc). These can be either global, explaining model behaviour in general (eg. PDPs, Functional Decomposition, Mechanistic Interpretability etc), or local, explaining individual predictions made by a model (eg. ICE, LIME, anchors, counterfactuals). Some model explanation methods are specific to certain kinds of models (eg. TCAV, Saliency Maps, gini-importance), whereas others are considered to be model agnostic (counterfactual explanations, LIME, anchors, etc). SHAP is a widely used model-agnostic, post-hoc, local explanation method, that produces scores indicating the relative importance of various input features that led to a model prediction²¹. SHAP is often the first technique used by practitioners interested in explaining the behaviour of black box models.

A.2 Explainability in Experimentation and Production

One use of explainability in machine learning is to continuously explain the predictions being made by artificial intelligence systems that have been deployed in the real world. This can help debug errors³⁹, comply with regulations, and engender trust⁴³. Often, explanations are used to gauge if protected attributes are significantly impacting model outputs and bring transparency to otherwise opaque decision making processes⁵. When models are being deployed in real world settings and affecting large numbers of users, it can be important to monitor their behaviour and have *explainability in production*, which requires explanations to be computed in a scalable manner.

A different use of explainability is in the lab, where researchers may want to understand model behaviour in order to study underlying phenomena. When using explainability during model development or exploratory data analysis, explanations can be used to understand the data generating process itself and building machine learning models and then using explainability techniques such as SHAP has been proposed as a means of scientific discovery^{6,7,9,13,33}. This strategy has been employed in diverse fields such as policy-making¹, physics⁴, engineering^{15,38}, epidemiology^{3,42}, chemistry^{2,11}, healthcare⁸, ecology¹⁴, and geology²⁹, research where model explanations are used to derive insights about underlying natural phenomena using a machine learning model, rather than explain the behaviour of a model used in decision making. This can be termed *explainability in experimentation*, and requires explanations to be robust and consistent.

These two use cases for model explanations are different, and present themselves with different desiderata and technical challenges. ROT provides a scalable method that is important in production settings, as well as a robust method that does not suffer from causal vulnerabilities or the Rashomon effect important in experimentation settings.

A.3 Implicit Assumptions about Causality

Machine learning classification is concerned with learning the conditional probability of obtaining a response y from a given set of inputs x . Borrowing from Judea Pearl's causal hierarchy²⁷, this involves "level 1" reasoning - the ability to observe associations. There is broad consensus today that machine learning alone is unable to produce systems with higher "level 2" or "level 3" reasoning abilities involving the ability to answer "interventional" and "counterfactual" questions, respectively.²⁷

The AI explainability literature so far has been surprisingly forgiving of this deficit typical in all modern machine learning systems. Sensitivity based model explanation methods like SHAP implicitly assume that models produce predictions using level 3 reasoning. If a model only possesses "level 1" observational reasoning, but an explanation method probes it with inputs that do not exist in naturally observed training data, then model predictions in these regions cannot be considered indicative of model behaviours.

SHAP's reliance on perturbing model inputs causes explanations to be overly sensitive to out of distribution model behaviour, which can be arbitrary¹⁹. The solutions proposed to this problem such as manifold-SHAP⁴⁰ or causal SHAP¹⁶ necessarily involve solving the harder problems of density estimation⁴¹ or causal modeling²⁷, as opposed to learning a simpler classifier decision boundary. Adversarial attacks have been proposed to exploit this vulnerability, and it has been shown that explanations can be manipulated this way, leading to fair-washing.³⁶

While these two approaches are possible to use in theory, the associated problems with each often make them impossible to implement in practical scenarios. These problems are especially pronounced for uses of model explainability in experimentation, where many models are built and explained iteratively, and the focus is more on the explanations themselves than on deploying the model to real world production settings. In such scenarios, both requiring the user to first build a causal model or estimate the entire density of the data is a non-starter.

By contrast, ROT makes no causal assumptions about the model, and does not need to solve the “harder” problems of density estimation either. ROT generates explanations entirely through “level 1” observation.

A.4 Ablation Studies and Sensitivity Analysis

Feature importances can be motivated in at least two distinct ways, involving the effect of:

1. removing or ablating input features on model outputs to determine feature importances through their predictiveness (**ablation studies**); or
2. perturbing input features and measuring the corresponding perturbations on model outputs to determine feature importances through their impact on the model output (**sensitivity analysis**).

Both these notions have been used widely. Examples of sensitivity analysis include applications in chemistry³⁵³², geology²³²⁰ and engineering²⁸. Ablation studies have roots in medicine and are widely used in machine learning²⁴ today, and sometimes in other fields like physics³¹¹⁷ and biology³⁴²⁶.

	Global Explanations for Overall Model Behaviour	Local Explanations for Individual Predictions
Feature Sensitivity Determines Feature Importance (eg. Sensitivity Analysis)	1 Partial Dependence Plots 2 Accumulated Local Effects 3 ⋮	1 LIME & SHAP 2 Counterfactuals 3 ⋮
Feature Predictiveness Determines Feature Importance (eg. Ablation Studies)	1 Permutation Feature Importance 2 MDI (for tree ensembles) 3 ⋮	ROT

Table 4. Among those model explanation methods that use feature predictiveness to determine importance, ROT is the only local model explanation method. Amongst all local model explanation methods, ROT is the only method that does not use feature sensitivity to determine feature importance.

In existing machine learning explainability literature, there exist global methods motivated by sensitivity analysis, local methods motivated by sensitivity analysis, global methods motivated by ablation studies, but no local methods motivated by ablation studies. ROT is the first model agnostic post-hoc local explanation method that defines feature importance using the predictiveness of features rather than the sensitivity of the output to perturbations, akin to ablation studies rather than sensitivity analysis. Table 4 provides a non-exhaustive list of model explanation methods, classified along different axes: whether the explanations are global or local, and whether feature importance is determined by sensitivity or predictiveness. ROT is a unique explanation method, as can be seen from the table.

A.5 Problems with Model Explanations in Practice

Returning to our previous examples of trying to debug LLM behaviour by explaining its classification outputs, and of trying to monitor a large-scale proprietary e-commerce recommendation system, we can illustrate the common practical problems explaining machine learning predictions.

When trying to explain the behaviour of an LLM, it is not possible to arbitrarily modify the input features to the model. For an explanation method like SHAP, it then becomes unclear how to perturb the inputs features. Additionally, explanations can only be used to ascribe importances to input features, and nuanced analyses studying the relationship between the output and external concepts or latent features is difficult. ROT is useful because not only does it not rely on input perturbations, but also because it allows the inclusion of external concepts when generating model explanations. TCAV is a global explanation method that explicitly uses latent concepts to provide model explanations¹⁸, but ROT is able to do this on a local, per-prediction level.

When trying to audit the behaviour of proprietary systems, often model artifacts cannot be accessed to obtain output predictions for new input points. This makes explanation methods like SHAP impossible to use without first creating a model to mimic the original. In real world settings, it is often easier to explain such a mimic model, rather than explaining the actual model¹⁰¹², and it is hard to assert that the mimic explanations generated also apply to the original model due the predictive multiplicity of classifiers²². This “Rashomon effect”, where two models have similar predictions but different internal

809 mechanisms³⁰, makes model explanations obtained using a mimic model inherently untrustworthy when explaining the original
810 model behaviour²⁵. ROT sidesteps this issue entirely and does not need access to a model at all, it generates model explanations
811 instead by looking at its predictions.

B Derivation

We wish to explain the behavior of a model $C(\cdot)$, over an empiric data distribution X . Given a data-point, $x \in X$, we use x_j to refer to the j^{th} component of the feature vector of x . We will define our explanations as a set of functions, similarly indexed by j , and refer to $f_j(x_j)$ as the importance of x_j . Given an arbitrary subset of features $J \subseteq \mathbb{J}$ over a data-point x , we wish to be able make predictions about the model response $C(x)$ using only the sum of importance weights $\sum_{j \in J} f_j(x_j)$

In this way, we wish to be able to make a prediction about the model response $C(x)$ using incomplete information about the feature vector of x , where only $j \in J$ of the feature values are known. The $f_j(x_j)$ individually serve as feature importances, and when summed are used to predict the model output.

B.1 Formulating Explanations from Incomplete Information

We give a general formulation suitable both for modelling discrete decisions using a logistic regression approximation, and for modelling continuous regression tasks using a linear regression approximation. In practice, these can be replaced with any additive model.

The functions f_j are parameterized by weights θ_j and the model is approximated by a function F of their sum and a global bias term G . Recall that we want to find the optimal set of additive functions f such that for any data point $x \in X$ and arbitrary subset of features J we can approximate $C(x)$.

$$C(x) \approx F\left(\sum_{j \in J} f_{\theta_j}(x_j) + G\right) \quad \forall J \in \mathcal{P}(\mathbb{J}) \quad (6)$$

As mentioned above, we use the linear and logistic approximations of $C(\cdot)$, in which $f_{\theta_j}(x_j) = a_j x_j + b_j$. In the case of linear regression F is the identity function, whereas for logistic regression F is the logistic function σ .

For regression models, this yields:

$$C(x) \approx \sum_{j \in J} (a_j x_j + b_j) + G \quad \forall J \in \mathcal{P}(\mathbb{J}) \quad (7)$$

and for classification this yields:

$$C(x) \approx \sigma\left(\sum_{j \in J} (a_j x_j + b_j) + G\right) \quad \forall J \in \mathcal{P}(\mathbb{J}) \quad (8)$$

For regression models, we minimise the mean squared error, further simplifying equation 7. The per feature bias term b_j can be disregarded if the mean of every feature j is 0, and the global bias term becomes equal to the mean model response $\bar{C}(x)$.

For classification models, we maximize the likelihood by marginalising over a distribution of randomly sampled features. We seek the values of θ_j that minimize the expected loss $\mathbb{L}[\cdot, \cdot]$ (typically cross-entropy error) over all subsets of features $J \subseteq \mathbb{J}$ and data-points $x \in X$.

$$\min \sum_{J \subseteq \mathbb{J}} \left(p^{|J|} (1-p)^{(|\mathbb{J}|-|J|)} \cdot \sum_{x \in X} \mathbb{L}\left[C(x), \sigma\left(\sum_{j \in J} f_{\theta_j}(x_j) + G\right)\right] \right) \quad (9)$$

Like the formulation of SHAP, this objective is exponentially large with respect to the feature space. However, a key difference that allows us solve it effectively is that the number of function calls to C is equal to the number of data points $|X|$, and completely independent of the feature space J . By commuting this is equivalent to

$$\min \sum_{x \in X} \sum_{J \subseteq \mathbb{J}} \left(p^{|J|} (1-p)^{(|\mathbb{J}|-|J|)} \cdot \mathbb{L}\left[C(x), \sigma\left(\sum_{j \in J} f_{\theta_j}(x_j) + G\right)\right] \right) \quad (10)$$

Upon careful inspection, this is equivalent to performing dropout with probability p^{37} on the initial approximation model $\sigma\left(\sum_{j \in J} f_{\theta_j}(x_j) + G\right)$. This allows us to directly find the weights θ_j (recall we defined $f_{\theta_j} = a_j x_j + b_j$) and G using stochastic optimization by treating this as a regression problem regulated by dropout. Implementation of the method is straightforward, with the only subtlety to be aware of is that the weights and biases should be zero-initialized to avoid problems with vanishing gradients in the logistic regression loss.

844 **B.2 Uniqueness and Stability**

845 Each component of the objective defined in equation (10) is convex. As such, if for every estimated variable there exists a
846 strictly convex component containing that variable, the problem as a whole is strictly convex. Therefore, the overall objective
847 has a unique minimum provided the individual functions $f_{\theta_j}(\cdot)$ are well-posed and strictly convex with each function having a
848 single minimum associated with it.

849 **B.3 From Approximation Model Weights to Interpretable Feature Attributions**

850 The optimization procedure returns a set of functions f_{θ_j} that then need to be applied to each feature. The response $f_{\theta_j}(x_j)$
851 can be directly interpreted as a signed measure of the importance of feature j on data-point x . For logistic regression, the sum
852 $\sum_{j \in J} f_{\theta_j}(x_j)$ can be directly interpreted as probability in logit space of the classifier response $C(x)$; while for linear regression
853 the same sum is an estimate of the model prediction $C(x)$.

C Extended Results

C.1 Evaluating Zero-Shot LLM Classifier Models with XAI

Explaining Judicial Case Predictions from a Fine-Tuned RoBERTa Model Example case from our experiment explaining LLM predictions of judicial case outcomes.

Kapur, J. 1. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is the assessee and the respondent is the Commissioner of Income-tax and the question raised is as to the applicability of S. 10(2)(xv) of the Indian Income-tax Act to a gratuity paid by the appellant to one of its officers on his retirement from service. 2. The appeal relates to the assessment year 1950-51. M/s. Gordon Woodroffee and Co. (Madras) Ltd., was incorporated as a private limited company in 1922 and became the Managing Agent of a public limited company M/s. Gordon Woodroffee Leather Manufacturing Company Ltd., which is the assessee. One J. H. Philips was employed in the Managing Agent Company from 1922 to 1935 and from 1935 he became an employee of the appellant company and became its Director from 1940. On March 22, 1949, he wrote a letter to the appellant company expressing his intention to resign from the Board of the Company as from April 4, 1949 upon his retirement from the employment of the company and requested that his resignation be accepted. On March 24, 1949, the Board of Directors of the appellant Company passed a resolution that his resignation be accepted and in appreciation of his long and valuable services to the Company he be paid a gratuity of Rs. 50,000/- out of which the appellant Company was to pay Rs. 40,000/- and the Managing Agent Company the balance of Rs. 10,000/-. On April 4, 1949 this resolution of the Board of Directors was confirmed, on the same date a resolution to the same effect was passed at an Extraordinary General Meeting of the Company and before the end of its accounting year i.e., October 31, 1949, this amount of Rs. 40,000/- was paid to Mr. J. H. Philips. 3. This amount was claimed as a deduction under S. 10(2)(xv) of the Income-tax Act which reads :-Section 10 (2) : ""Such profits or gains shall be computed after making the following allowances, namely :
..... (xv) any expenditure (not being an allowance of the nature described in any of the clauses (i) to (xiv) inclusive, and not being in the nature of capital expenditure or personal expenses of the assessee) laid out or expended wholly and exclusively for the purpose of such business, profession or vocation."" The amount was disallowed by the Income-tax Officer as well as by the Appellate Assistant Commissioner on the ground that the appellant Company had no pension scheme; the payment was voluntary and that the entry in the assessee's books clearly indicated it to be a capital payment. Against this order the appellant Company took an appeal to the Income-tax Appellate Tribunal which upheld the order of the Appellate Assistant Commissioner. It held that according to the resolution the gratuity was paid ""for long and valuable services to the Company""; that there was nothing to indicate that Mr. J. H. Philips had accepted a lower salary in expectation of getting a gratuity at the end of his service; at there was no such practice in the appellant Company and that during the course of his service he was being remunerated at a graduated scale of salary and a commission of it 2 1/2 on the profits; that there was no ""expectancy"" that at the end of the service there would be a recompense for faithful and efficient service; that he had been suitably rewarded by being given a commission on the profits ""in order to whip up his enthusiasm"". It was also mentioned that in the books of the appellant Company the amount had not been debited in the profit and loss account but was debited to the appropriation account thereby indicating that it was an extra payment or a payment made in the nature of a capital expense. Taking all these circumstances into consideration the Tribunal came to the conclusion that it was difficult to hold that the expenditure was not in the nature of a capital expenditure or that it was expended wholly exclusively for the purpose of the assessee's business. At the instance of the appellant Company the case was stated to the High Court under S. 66(1) of the Income-tax Act and the following question was referred :-""Whether the sum of Rs. 40,000/- paid to Mr. J. H. Philips on his retirement from the service of the Company was, not an admissible deduction under S. 10 (2)(xv) of the Income-tax Act, 1922."" 4. The High Court answered the question against the appellant Company. It held that in order that S. 10(2) (xv) applicable it had to be proved that the amount was laid out or expended wholly and exclusively for purposes of the company's business. In his case the amount was paid on retirement and for valuable services rendered by Mr. J. H. Philips; there was no evidence that he expected to receive this amount or the Company contemplated its payment at any time before; the payment was voluntary and there was no evidence to show that it was in the future interest of the business of the Company that the expenditure was incurred. The High Court observed : ""In the case of a payment of a gratuity to a retiring employee recognition of his past services, with nothing more cannot, in our opinion satisfy the requirements of S. 10(2) (xv), even if those requirements are judged from the view point of commercial expediency, as it always should be when a claim arises under S. 10(2)(xv). Was the expenditure incurred in the future interest of the business of the assessee? Was there any connection between the purpose of the payment and the further conduct of the business of the assessee? These are the tests to be satisfied before it could be said that in paying the gratuity money was laid out or expended wholly and exclusively for the purpose of the business of the company. These tests the assessee did not satisfy in this case,"" Against this judgment and order the appellant Company has brought this appeal by special leave. 5. It was argued on behalf of the appellant that the amount had been paid as a matter of commercial expediency and in the interest of the Company as an inducement to other employees that if they rendered service in a similar manner with efficiency and honesty they would be similarly rewarded. Decisive test, it was submitted was whether such payments of gratuity were likely in future also and was the payment made as an incentive to the employees to give their best to the employer and if it was so then the payment was a matter of commercial prudence. It was also submitted that the Company had acted not with any oblique motive and its good faith was not in doubt and in support of the contention several cases were relied upon,

6. In our opinion on the findings as given the payment in dispute does not fall within the provisions of S. 10(2)(xv). The amount was paid not in pursuance of any scheme of payment of gratuities nor was it an amount which the recipient expected to be paid for long and faithful service but it was a voluntary payment not with the object of facilitating the carrying on of the business of the appellant Company or as a matter of commercial expediency but in recognition of long and faithful service of Mr. J. H. Philips. There was no practice in the appellant Company to pay such amounts and it did not affect the quantum of salary of the recipient

. The two cases strongly relied upon by the appellant Company were J. P. Hancock v. General Reversionary and Investment Company Ltd. (1919) 7 Tax Cas 358 and J. W. Smith v. Incorporated Council of Law Reporting for England and Wales (1914) 6 Tax Cas 477. In the former case the assessee Company sought to charge as a trade expense a lump sum which it had paid for the purchase, for the benefit of a former actuary, of an annuity equal in amount to the pension which the Company had resolved to pay him. This was held to be an expense admissible in computing the Company's profits assessable to income-tax. But in that case it was the practice of the assessee company to grant pensions to its servants after a considerable period of service and this practice was known to the employees and affected the rate of salary paid by the Company in that the employees were willing to serve the Company at lower rates than they otherwise would have by reason of the expectation of the pension at the end of their service. In the latter case there was a practice of granting gratuities and that was the ground for holding the amount to be a proper deduction.

7. In our opinion the proper test to apply in this case is, was the payment made as a matter of practice which affected the quantum of salary or was there an expectation by the employee of getting a gratuity or was the sum of money expended on the ground of commercial expediency and in order indirectly to facilitate the carrying on of the business. But this has not been shown and therefore the amounts claimed is not a deductible item under S. 10(2)(xv).

Figure 9. Unabridged explanation for the case from Figure 2, where the appeal was rejected. The “ground-truth” importance annotation is indicated using a bold font, and the highlights are produced using RoT. RoT aligns well with the annotation.

Kapur, J. 1. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is the assessee and the respondent is the Commissioner of Income-tax and the question raised is as to the applicability of S. 10(2)(xv) of the Indian Income-tax Act to a gratuity paid by the appellant to one of its officers on his retirement from service. 2. The appeal relates to the assessment year 1950-51. M/s. Gordon Woodroffee and Co. (Madras) Ltd., was incorporated as a private limited company in 1922 and became the Managing Agent of a public limited company M/s. Gordon Woodroffee Leather Manufacturing Company Ltd., which is the assessee. One J. H. Philips was employed in the Managing Agent Company from 1922 to 1935 and from 1935 he became an employee of the appellant company and became its Director from 1940. On March 22, 1949, he wrote a letter to the appellant company expressing his intention to resign from the Board of the Company as from April 4, 1949 upon his retirement from the employment of the company and requested that his resignation be accepted. On March 24, 1949, the Board of Directors of the appellant Company passed a resolution that his resignation be accepted and in appreciation of his long and valuable services to the Company he be paid a gratuity of Rs. 50,000/- out of which the appellant Company was to pay Rs. 40,000/- and the Managing Agent Company the balance of Rs. 10,000/-. On April 4, 1949 this resolution of the Board of Directors was confirmed, on the same date a resolution to the same effect was passed at an Extraordinary General Meeting of the Company and before the end of its accounting year i.e., October 31, 1949, this amount of Rs. 40,000/- was paid to Mr. J. H. Philips. 3. This amount was claimed as a deduction under S. 10(2)(xv) of the Income-tax Act which reads :-Section 10 (2) : "Such profits or gains shall be computed after making the following allowances, namely :
..... (xv) any expenditure (not being an allowance of the nature described in any of the clauses (i) to (xiv) inclusive, and not being in the nature of capital expenditure or personal expenses of the assessee) laid out or expended wholly and exclusively for the purpose of such business, profession or vocation." The amount was disallowed by the Income-tax Officer as well as by the Appellate Assistant Commissioner on the ground that the appellant Company had no pension scheme; the payment was voluntary and that the entry in the assessee's books clearly indicated it to be a capital payment. Against this order the appellant Company took an appeal to the Income-tax Appellate Tribunal which upheld the order of the Appellate Assistant Commissioner. It held that according to the resolution the gratuity was paid "for long and valuable services to the Company"; that there was nothing to indicate that Mr. J. H. Philips had accepted a lower salary in expectation of getting a gratuity at the end of his service; at there was no such practice in the appellant Company and that during the course of his service he was being remunerated at a graduated scale of salary and a commission of it 2 1/2 on the profits; that there was no "expectancy" that at the end of the service there would be a recompense for faithful and efficient service; that he had been suitably rewarded by being given a commission on the profits "in order to whip up his enthusiasm". It was also mentioned that in the books of the appellant Company the amount had not been debited in the profit and loss account but was debited to the appropriation account thereby indicating that it was an extra payment or a payment made in the nature of a capital expense. Taking all these circumstances into consideration the Tribunal came to the conclusion that it was difficult to hold that the expenditure was not in the nature of a capital expenditure or that it was expended wholly exclusively for the purpose of the assessee's business. At the instance of the appellant Company the case was stated to the High Court under S. 66(1) of the Income-tax Act and the following question was referred :-"Whether the sum of Rs. 40,000/- paid to Mr. J. H. Philips on his retirement from the service of the Company was, not an admissible deduction under S. 10 (2)(xv) of the Income-tax Act, 1922." 4. The High Court answered the question against the appellant Company. It held that in order that S. 10(2) (xv) applicable it had to be proved that the amount was laid out or expended wholly and exclusively for purposes of the company's business. In his case the amount was paid on retirement and for valuable services rendered by Mr. J. H. Philips; there was no evidence that he expected to receive this amount or the Company contemplated its payment at any time before; the payment was voluntary and there was no evidence to show that it was in the future interest of the business of the Company that the expenditure was incurred. The High Court observed : "In the case of a payment of a gratuity to a retiring employee recognition of his past services, with nothing more cannot, in our opinion satisfy the requirements of S. 10(2) (xv), even if those requirements are judged from the view point of commercial expediency, as it always should be when a claim arises under S. 10(2)(xv). Was the expenditure incurred in the future interest of the business of the assessee? Was there any connection between the purpose of the payment and the further conduct of the business of the assessee? These are the tests to be satisfied before it could be said that in paying the gratuity money was laid out or expended wholly and exclusively for the purpose of the business of the company. These tests the assessee did not satisfy in this case." Against this judgment and order the appellant Company has brought this appeal by special leave. 5. It was argued on behalf of the appellant that the amount had been paid as a matter of commercial expediency and in the interest of the Company as an inducement to other employees that if they rendered service in a similar manner with efficiency and honesty they would be similarly rewarded. Decisive test, it was submitted was whether such payments of gratuity were likely in future also and was the payment made as an incentive to the employees to give their best to the employer and if it was so then the payment was a matter of commercial prudence. It was also submitted that the Company had acted not with any oblique motive and its good faith was not in doubt and in support of the contention several cases were relied upon,

6. In our opinion on the findings as given the payment in dispute does not fall within the provisions of S. 10(2)(xv). The amount was paid not in pursuance of any scheme of payment of gratuities nor was it an amount which the recipient expected to be paid for long and faithful service but it was a voluntary payment not with the object of facilitating the carrying on of the business of the appellant Company or as a matter of commercial expediency but in recognition of long and faithful service of Mr. J. H. Philips. There was no practice in the appellant Company to pay such amounts and it did not affect the quantum of salary of the recipient

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7. In our opinion the proper test to apply in this case is, was the payment made as a matter of practice which affected the quantum of salary or was there an expectation by the employee of getting a gratuity or was the sum of money expended on the ground of commercial expediency and in order indirectly to facilitate the carrying on of the business. But this has not been shown and therefore the amounts claimed is not a deductible item under S. 10(2)(xv).

Figure 10. Unabridged explanation for the case from Figure 2, where the appeal was rejected. The “ground-truth” importance annotation is indicated using a bold font, and the highlights are produced using SHAP. SHAP does not align as well with the annotation as RoT from Figure 9.

Kapur, J. 1. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is the assessee and the respondent is the Commissioner of Income-tax and the question raised is as to the applicability of S. 10(2)(xv) of the Indian Income-tax Act to a gratuity paid by the appellant to one of its officers on his retirement from service. 2. The appeal relates to the assessment year 1950-51. M/s. Gordon Woodroffee and Co. (Madras) Ltd., was incorporated as a private limited company in 1922 and became the Managing Agent of a public limited company M/s. Gordon Woodroffee Leather Manufacturing Company Ltd., which is the assessee. One J. H. Philips was employed in the Managing Agent Company from 1922 to 1935 and from 1935 he became an employee of the appellant company and became its Director from 1940. On March 22, 1949, he wrote a letter to the appellant company expressing his intention to resign from the Board of the Company as from April 4, 1949 upon his retirement from the employment of the company and requested that his resignation be accepted. On March 24, 1949, the Board of Directors of the appellant Company passed a resolution that his resignation be accepted and in appreciation of his long and valuable services to the Company he be paid a gratuity of Rs. 50,000/- out of which the appellant Company was to pay Rs. 40,000/- and the Managing Agent Company the balance of Rs. 10,000/-. On April 4, 1949 this resolution of the Board of Directors was confirmed, on the same date a resolution to the same effect was passed at an Extraordinary General Meeting of the Company and before the end of its accounting year i.e., October 31, 1949, this amount of Rs. 40,000/- was paid to Mr. J. H. Philips. 3. This amount was claimed as a deduction under S. 10(2)(xv) of the Income-tax Act which reads :-Section 10 (2) : "Such profits or gains shall be computed after making the following allowances, namely :
..... (xv) any expenditure (not being an allowance of the nature described in any of the clauses (i) to (xiv) inclusive, and not being in the nature of capital expenditure or personal expenses of the assessee) laid out or expended wholly and exclusively for the purpose of such business, profession or vocation." The amount was disallowed by the Income-tax Officer as well as by the Appellate Assistant Commissioner on the ground that the appellant Company had no pension scheme; the payment was voluntary and that the entry in the assessee's books clearly indicated it to be a capital payment. Against this order the appellant Company took an appeal to the Income-tax Appellate Tribunal which upheld the order of the Appellate Assistant Commissioner. It held that according to the resolution the gratuity was paid "for long and valuable services to the Company"; that there was nothing to indicate that Mr. J. H. Philips had accepted a lower salary in expectation of getting a gratuity at the end of his service; at there was no such practice in the appellant Company and that during the course of his service he was being remunerated at a graduated scale of salary and a commission of it 2 1/2 on the profits; that there was no "expectancy" that at the end of the service there would be a recompense for faithful and efficient service; that he had been suitably rewarded by being given a commission on the profits "in order to whip up his enthusiasm". It was also mentioned that in the books of the appellant Company the amount had not been debited in the profit and loss account but was debited to the appropriation account thereby indicating that it was an extra payment or a payment made in the nature of a capital expense. Taking all these circumstances into consideration the Tribunal came to the conclusion that it was difficult to hold that the expenditure was not in the nature of a capital expenditure or that it was expended wholly exclusively for the purpose of the assessee's business. At the instance of the appellant Company the case was stated to the High Court under S. 66(1) of the Income-tax Act and the following question was referred :-"Whether the sum of Rs. 40,000/- paid to Mr. J. H. Philips on his retirement from the service of the Company was, not an admissible deduction under S. 10 (2)(xv) of the Income-tax Act, 1922." 4. The High Court answered the question against the appellant Company. It held that in order that S. 10(2) (xv) applicable it had to be proved that the amount was laid out or expended wholly and exclusively for purposes of the company's business. In his case the amount was paid on retirement and for valuable services rendered by Mr. J. H. Philips; there was no evidence that he expected to receive this amount or the Company contemplated its payment at any time before; the payment was voluntary and there was no evidence to show that it was in the future interest of the business of the Company that the expenditure was incurred. The High Court observed : "In the case of a payment of a gratuity to a retiring employee recognition of his past services, with nothing more cannot, in our opinion satisfy the requirements of S. 10(2) (xv), even if those requirements are judged from the view point of commercial expediency, as it always should be when a claim arises under S. 10(2)(xv). Was the expenditure incurred in the future interest of the business of the assessee? Was there any connection between the purpose of the payment and the further conduct of the business of the assessee? These are the tests to be satisfied before it could be said that in paying the gratuity money was laid out or expended wholly and exclusively for the purpose of the business of the company. These tests the assessee did not satisfy in this case," Against this judgment and order the appellant Company has brought this appeal by special leave. 5. It was argued on behalf of the appellant that the amount had been paid as a matter of commercial expediency and in the interest of the Company as an inducement to other employees that if they rendered service in a similar manner with efficiency and honesty they would be similarly rewarded. Decisive test, it was submitted was whether such payments of gratuity were likely in future also and was the payment made as an incentive to the employees to give their best to the employer and if it was so then the payment was a matter of commercial prudence. It was also submitted that the Company had acted not with any oblique motive and its good faith was not in doubt and in support of the contention several cases were relied upon,

6. In our opinion on the findings as given the payment in dispute does not fall within the provisions of S. 10(2)(xv). The amount was paid not in pursuance of any scheme of payment of gratuities nor was it an amount which the recipient expected to be paid for long and faithful service but it was a voluntary payment not with the object of facilitating the carrying on of the business of the appellant Company or as a matter of commercial expediency but in recognition of long and faithful service of Mr. J. H. Philips. There was no practice in the appellant Company to pay such amounts and it did not affect the quantum of salary of the recipient

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Figure 11. Explanation from additional explainer for the case from Figure 2, where the appeal was rejected. The “ground-truth” importance annotation is indicated using a bold font, and the highlights are produced using Integrated Gradients.

Kapur, J. 1. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is the assessee and the respondent is the Commissioner of Income-tax and the question raised is as to the applicability of S. 10(2)(xv) of the Indian Income-tax Act to a gratuity paid by the appellant to one of its officers on his retirement from service. 2. The appeal relates to the assessment year 1950-51. M/s. Gordon Woodroffee and Co. (Madras) Ltd., was incorporated as a private limited company in 1922 and became the Managing Agent of a public limited company M/s. Gordon Woodroffee Leather Manufacturing Company Ltd., which is the assessee. One J. H. Philips was employed in the Managing Agent Company from 1922 to 1935 and from 1935 he became an employee of the appellant company and became its Director from 1940. On March 22, 1949, he wrote a letter to the appellant company expressing his intention to resign from the Board of the Company as from April 4, 1949 upon his retirement from the employment of the company and requested that his resignation be accepted. On March 24, 1949, the Board of Directors of the appellant Company passed a resolution that his resignation be accepted and in appreciation of his long and valuable services to the Company he be paid a gratuity of Rs. 50,000/- out of which the appellant Company was to pay Rs. 40,000/- and the Managing Agent Company the balance of Rs. 10,000/-. On April 4, 1949 this resolution of the Board of Directors was confirmed, on the same date a resolution to the same effect was passed at an Extraordinary General Meeting of the Company and before the end of its accounting year i.e., October 31, 1949, this amount of Rs. 40,000/- was paid to Mr. J. H. Philips. 3. This amount was claimed as a deduction under S. 10(2)(xv) of the Income-tax Act which reads :-Section 10 (2) : "Such profits or gains shall be computed after making the following allowances, namely :
..... (xv) any expenditure (not being an allowance of the nature described in any of the clauses (i) to (xiv) inclusive, and not being in the nature of capital expenditure or personal expenses of the assessee) laid out or expended wholly and exclusively for the purpose of such business, profession or vocation." The amount was disallowed by the Income-tax Officer as well as by the Appellate Assistant Commissioner on the ground that the appellant Company had no pension scheme; the payment was voluntary and that the entry in the assessee's books clearly indicated it to be a capital payment. Against this order the appellant Company took an appeal to the Income-tax Appellate Tribunal which upheld the order of the Appellate Assistant Commissioner. It held that according to the resolution the gratuity was paid "for long and valuable services to the Company"; that there was nothing to indicate that Mr. J. H. Philips had accepted a lower salary in expectation of getting a gratuity at the end of his service; at there was no such practice in the appellant Company and that during the course of his service he was being remunerated at a graduated scale of salary and a commission of it 2 1/2 on the profits; that there was no "expectancy" that at the end of the service there would be a recompense for faithful and efficient service; that he had been suitably rewarded by being given a commission on the profits "in order to whip up his enthusiasm". It was also mentioned that in the books of the appellant Company the amount had not been debited in the profit and loss account but was debited to the appropriation account thereby indicating that it was an extra payment or a payment made in the nature of a capital expense. Taking all these circumstances into consideration the Tribunal came to the conclusion that it was difficult to hold that the expenditure was not in the nature of a capital expenditure or that it was expended wholly exclusively for the purpose of the assessee's business. At the instance of the appellant Company the case was stated to the High Court under S. 66(1) of the Income-tax Act and the following question was referred :-"Whether the sum of Rs. 40,000/- paid to Mr. J. H. Philips on his retirement from the service of the Company was, not an admissible deduction under S. 10 (2)(xv) of the Income-tax Act, 1922." 4. The High Court answered the question against the appellant Company. It held that in order that S. 10(2) (xv) applicable it had to be proved that the amount was laid out or expended wholly and exclusively for purposes of the company's business. In his case the amount was paid on retirement and for valuable services rendered by Mr. J. H. Philips; there was no evidence that he expected to receive this amount or the Company contemplated its payment at any time before; the payment was voluntary and there was no evidence to show that it was in the future interest of the business of the Company that the expenditure was incurred. The High Court observed : "In the case of a payment of a gratuity to a retiring employee recognition of his past services, with nothing more cannot, in our opinion satisfy the requirements of S. 10(2) (xv), even if those requirements are judged from the view point of commercial expediency, as it always should be when a claim arises under S. 10(2)(xv). Was the expenditure incurred in the future interest of the business of the assessee? Was there any connection between the purpose of the payment and the further conduct of the business of the assessee? These are the tests to be satisfied before it could be said that in paying the gratuity money was laid out or expended wholly and exclusively for the purpose of the business of the company. These tests the assessee did not satisfy in this case." Against this judgment and order the appellant Company has brought this appeal by special leave. 5. It was argued on behalf of the appellant that the amount had been paid as a matter of commercial expediency and in the interest of the Company as an inducement to other employees that if they rendered service in a similar manner with efficiency and honesty they would be similarly rewarded. Decisive test, it was submitted was whether such payments of gratuity were likely in future also and was the payment made as an incentive to the employees to give their best to the employer and if it was so then the payment was a matter of commercial prudence. It was also submitted that the Company had acted not with any oblique motive and its good faith was not in doubt and in support of the contention several cases were relied upon,

6. In our opinion on the findings as given the payment in dispute does not fall within the provisions of S. 10(2)(xv). The amount was paid not in pursuance of any scheme of payment of gratuities nor was it an amount which the recipient expected to be paid for long and faithful service but it was a voluntary payment not with the object of facilitating the carrying on of the business of the appellant Company or as a matter of commercial expediency but in recognition of long and faithful service of Mr. J. H. Philips. There was no practice in the appellant Company to pay such amounts and it did not affect the quantum of salary of the recipient

. The two cases strongly relied upon by the appellant Company were J. P. Hancock v. General Reversionary and Investment Company Ltd. (1919) 7 Tax Cas 358 and J. W. Smith v. Incorporated Council of Law Reporting for England and Wales (1914) 6 Tax Cas 477. In the former case the assessee Company sought to charge as a trade expense a lump sum which it had paid for the purchase, for the benefit of a former actuary, of an annuity equal in amount to the pension which the Company had resolved to pay him. This was held to be an expense admissible in computing the Company's profits assessable to income-tax. But in that case it was the practice of the assessee company to grant pensions to its servants after a considerable period of service and this practice was known to the employees and affected the rate of salary paid by the Company in that the employees were willing to serve the Company at lower rates than they otherwise would have by reason of the expectation of the pension at the end of their service. In the latter case there was a practice of granting gratuities and that was the ground for holding the amount to be a proper deduction.

7. In our opinion the proper test to apply in this case is, was the payment made as a matter of practice which affected the quantum of salary or was there an expectation by the employee of getting a gratuity or was the sum of money expended on the ground of commercial expediency and in order indirectly to facilitate the carrying on of the business. But this has not been shown and therefore the amounts claimed is not a deductible item under S. 10(2)(xv).

Figure 12. Explanation from additional explainer for the case from Figure 2, where the appeal was rejected. The “ground-truth” importance annotation is indicated using a bold font, and the highlights are produced using LIME with 500 background samples (reduced from 5000).

Kapur, J. 1. This appeal by special leave is directed against the judgment and order of the High Court of Judicature at Madras. The appellant is the assessee and the respondent is the Commissioner of Income-tax and the question raised is as to the applicability of S. 10(2)(xv) of the Indian Income-tax Act to a gratuity paid by the appellant to one of its officers on his retirement from service. 2. The appeal relates to the assessment year 1950-51. M/s. Gordon Woodroffee and Co. (Madras) Ltd., was incorporated as a private limited company in 1922 and became the Managing Agent of a public limited company M/s. Gordon Woodroffee Leather Manufacturing Company Ltd., which is the assessee. One J. H. Philips was employed in the Managing Agent Company from 1922 to 1935 and from 1935 he became an employee of the appellant company and became its Director from 1940. On March 22, 1949, he wrote a letter to the appellant company expressing his intention to resign from the Board of the Company as from April 4, 1949 upon his retirement from the employment of the company and requested that his resignation be accepted. On March 24, 1949, the Board of Directors of the appellant Company passed a resolution that his resignation be accepted and in appreciation of his long and valuable services to the Company he be paid a gratuity of Rs. 50,000/- out of which the appellant Company was to pay Rs. 40,000/- and the Managing Agent Company the balance of Rs. 10,000/-. On April 4, 1949 this resolution of the Board of Directors was confirmed, on the same date a resolution to the same effect was passed at an Extraordinary General Meeting of the Company and before the end of its accounting year i.e., October 31, 1949, this amount of Rs. 40,000/- was paid to Mr. J. H. Philips. 3. This amount was claimed as a deduction under S. 10(2)(xv) of the Income-tax Act which reads :-Section 10 (2) : "Such profits or gains shall be computed after making the following allowances, namely :
..... (xv) any expenditure (not being an allowance of the nature described in any of the clauses (i) to (xiv) inclusive, and not being in the nature of capital expenditure or personal expenses of the assessee) laid out or expended wholly and exclusively for the purpose of such business, profession or vocation." The amount was disallowed by the Income-tax Officer as well as by the Appellate Assistant Commissioner on the ground that the appellant Company had no pension scheme; the payment was voluntary and that the entry in the assessee's books clearly indicated it to be a capital payment. Against this order the appellant Company took an appeal to the Income-tax Appellate Tribunal which upheld the order of the Appellate Assistant Commissioner. It held that according to the resolution the gratuity was paid "for long and valuable services to the Company"; that there was nothing to indicate that Mr. J. H. Philips had accepted a lower salary in expectation of getting a gratuity at the end of his service; at there was no such practice in the appellant Company and that during the course of his service he was being remunerated at a graduated scale of salary and a commission of it 2 1/2 on the profits; that there was no "expectancy" that at the end of the service there would be a recompense for faithful and efficient service; that he had been suitably rewarded by being given a commission on the profits "in order to whip up his enthusiasm". It was also mentioned that in the books of the appellant Company the amount had not been debited in the profit and loss account but was debited to the appropriation account thereby indicating that it was an extra payment or a payment made in the nature of a capital expense. Taking all these circumstances into consideration the Tribunal came to the conclusion that it was difficult to hold that the expenditure was not in the nature of a capital expenditure or that it was expended wholly exclusively for the purpose of the assessee's business. At the instance of the appellant Company the case was stated to the High Court under S. 66(1) of the Income-tax Act and the following question was referred :-"Whether the sum of Rs. 40,000/- paid to Mr. J. H. Philips on his retirement from the service of the Company was, not an admissible deduction under S. 10 (2)(xv) of the Income-tax Act, 1922." 4. The High Court answered the question against the appellant Company. It held that in order that S. 10(2) (xv) applicable it had to be proved that the amount was laid out or expended wholly and exclusively for purposes of the company's business. In his case the amount was paid on retirement and for valuable services rendered by Mr. J. H. Philips; there was no evidence that he expected to receive this amount or the Company contemplated its payment at any time before; the payment was voluntary and there was no evidence to show that it was in the future interest of the business of the Company that the expenditure was incurred. The High Court observed : "In the case of a payment of a gratuity to a retiring employee recognition of his past services, with nothing more cannot, in our opinion satisfy the requirements of S. 10(2) (xv), even if those requirements are judged from the view point of commercial expediency, as it always should be when a claim arises under S. 10(2)(xv). Was the expenditure incurred in the future interest of the business of the assessee? Was there any connection between the purpose of the payment and the further conduct of the business of the assessee? These are the tests to be satisfied before it could be said that in paying the gratuity money was laid out or expended wholly and exclusively for the purpose of the business of the company. These tests the assessee did not satisfy in this case." Against this judgment and order the appellant Company has brought this appeal by special leave. 5. It was argued on behalf of the appellant that the amount had been paid as a matter of commercial expediency and in the interest of the Company as an inducement to other employees that if they rendered service in a similar manner with efficiency and honesty they would be similarly rewarded. Decisive test, it was submitted was whether such payments of gratuity were likely in future also and was the payment made as an incentive to the employees to give their best to the employer and if it was so then the payment was a matter of commercial prudence. It was also submitted that the Company had acted not with any oblique motive and its good faith was not in doubt and in support of the contention several cases were relied upon,

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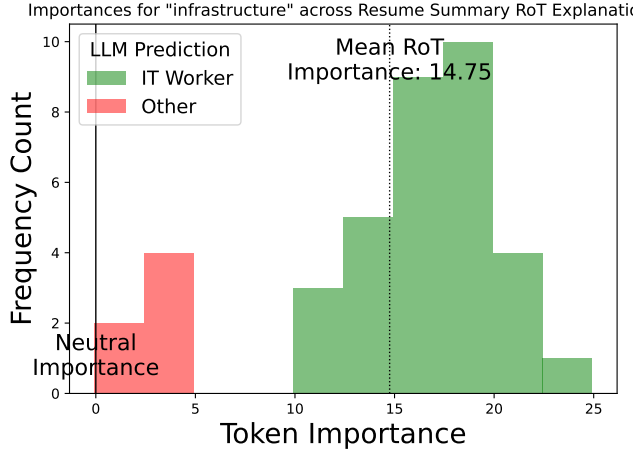
Figure 13. Explanation from additional explainer for the case from Figure 2, where the appeal was rejected. The “ground-truth” importance annotation is indicated using a bold font, and the highlights are produced using LIME with 5000 background samples (default configuration).

858 As documented in the **METHODS** section, the area under the receiver-operating-characteristic curve is more appropriate than
859 the area under the precision recall curve to evaluate LLM text explanations. However we report the PR-AUC for this experiment
860 in Table 5 for completeness. Further, we provide visualisations of additional case explanations at <https://anonymous.4open.science/w/Rule-of-Thumb-Explaining-Artificial-Intelligence-Systems-using-Partial-Information/JudicialCaseOutcomePrediction/Code/results.html>.
862

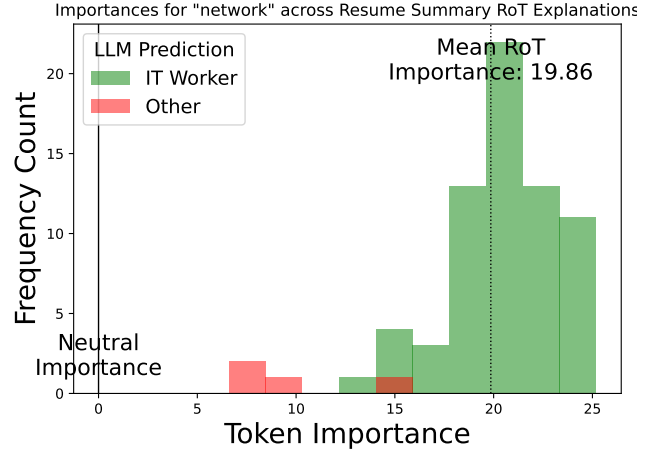
	RoT	SHAP (500 samples)	Integrated Gradients	LIME (500 samples)	Random
Average PR-AUC	<i>0.77251</i>	0.68255	0.75770	0.65712	0.53480

Table 5. Area under the precision-recall curve when comparing RoT explanations with ground-truth annotations. The underlying datapoints for this table are identical to Table 1 where we initially presented results from our experiment explaining the judicial case prediction AI system.

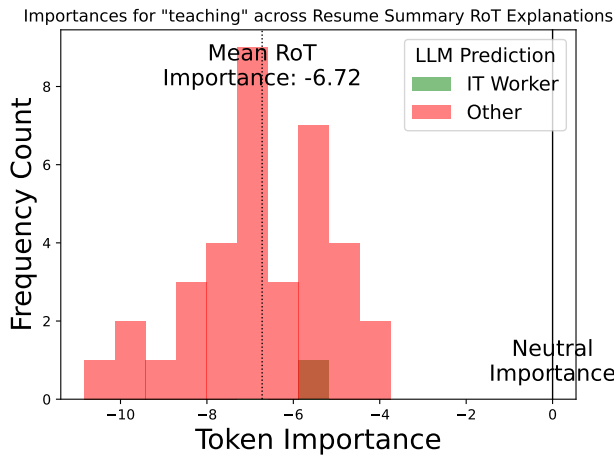
863 **Explaining Resume Filtering using Proprietary LLM APIs** . In addition to the wordclouds in figure 3, we can also analyse
864 how the importance of salient terms varies across resumes in the dataset, as in figure 14.



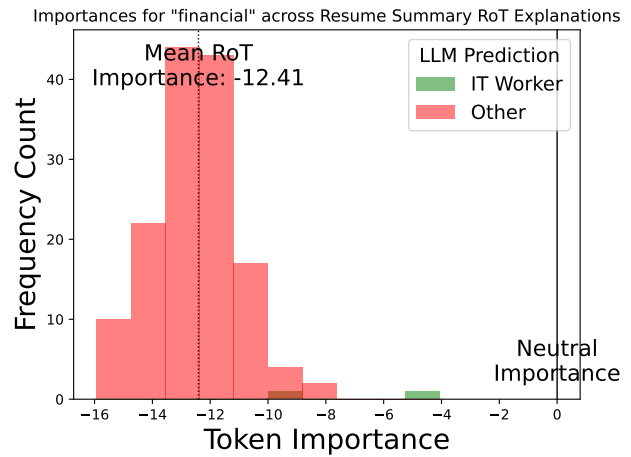
(a) RoT importances for “infrastructure” across our resume corpus



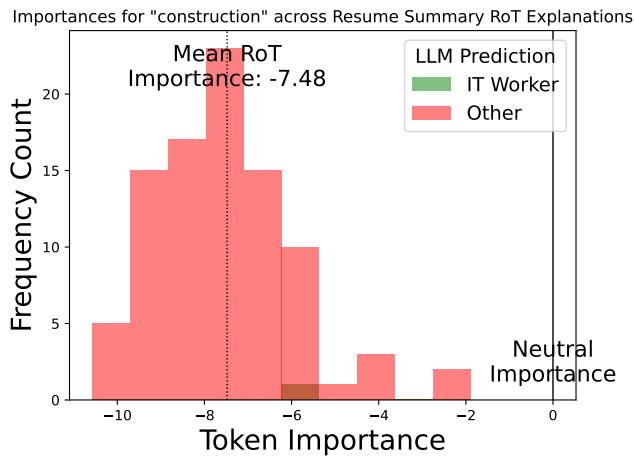
(b) RoT importances for “network” across our resume corpus



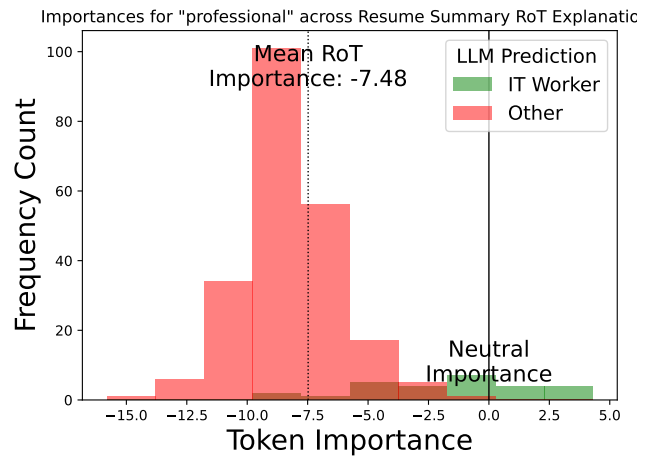
(c) RoT importances for “teaching” across our resume corpus



(d) RoT importances for “financial” across our resume corpus



(e) RoT importances for “construction” across our resume corpus



(f) RoT importances for “professional” across our resume corpus

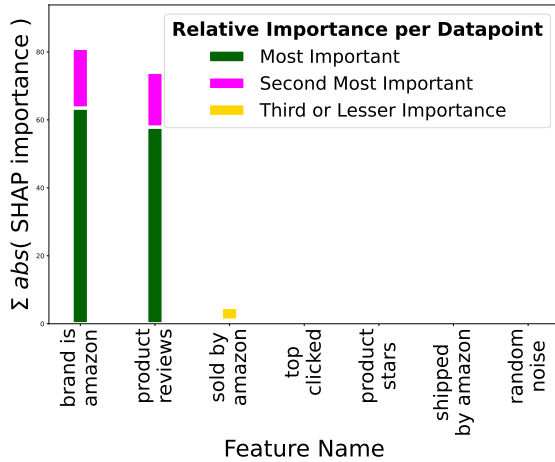
Figure 14. Histograms showing the importances obtained using RoT for “infrastructure”, “network”, “teaching”, “financial”, “construction”, and “professional” across resumes. The stacked bar plots are coloured green for token occurrences where the resume belonged to an IT worker, and red for others. Positive numbers on the X axis indicate RoT importances towards hiring as IT workers, and negative numbers indicate non IT workers.

865 C.2 Auditing Proprietary Black-Box AI Systems without Mimic Models

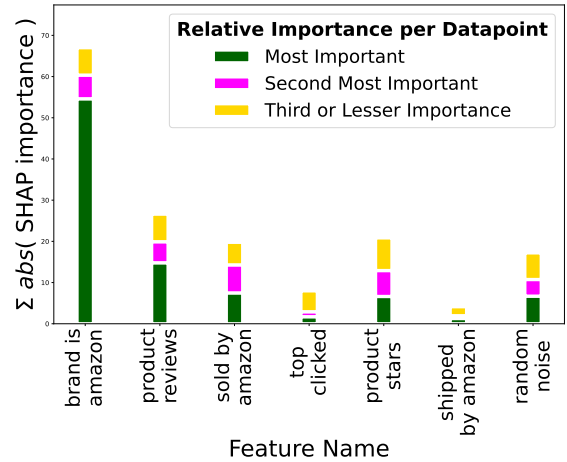
866 **Replicating The Markup’s audit of Amazon’s Product Recommendation Model** Using SHAP on a mimic model is not
 867 guaranteed to recover explanations faithful to the original model being audited. We constructed different “mimic” models that
 868 emulated the Amazon product recommendation system, and showed that SHAP explanations contradicted each other for the
 869 Markup created Random Forest, and the default hyper-parameter Random Forest. We now extend this analysis to the other
 870 logistic regression mimic models we created, producing extended versions of Figure 4 showing SHAP explanation distributions
 871 and SHAP explanation feature rankings respectively in figures 17 and 15 respectively. We also show global importances for all
 872 models, verifying that mimic models do indeed have different internal mechanisms despite producing predictions with similar
 873 accuracy rates.

	RF (Markup)	RF (Default)	LogReg	LogReg (L1)	LogReg (L2)
product stars	0.021	0.122	0.068	0.0	0.084
product reviews	0.055	0.246	0.042	0.0	0.032
brand is amazon	0.853	0.255	0.650	1.0	0.587
shipped by amazon	0.006	0.027	0.003	0.0	0.012
sold by amazon	0.052	0.082	0.177	0.0	0.245
top clicked	0.002	0.034	0.032	0.0	0.022
random noise	0.010	0.233	0.027	0.0	0.018

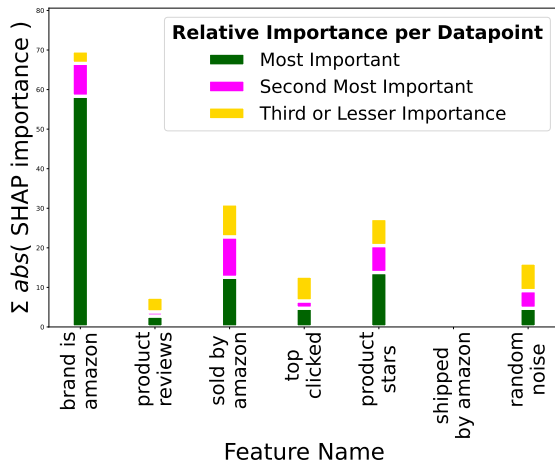
Table 6. Global feature importances (normalised) for the mimic models from Table 2. The two most important features per model are marked in **bold**. These global explanations (Mean Decrease in Impurity for RF, model coefficients for LogReg) vary significantly, showing that models in the Rashomon set make the same prediction through different internal mechanisms.



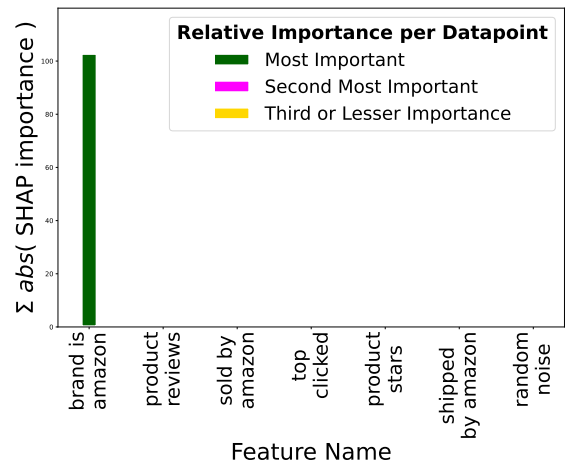
(a) The Markup model finds the most important feature to usually be either “brand is amazon” or “product reviews”



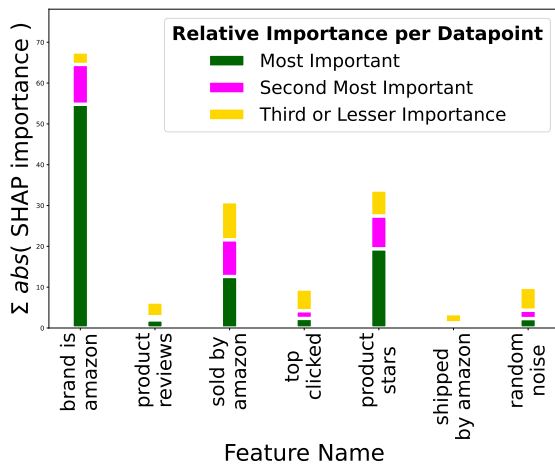
(b) The default random forest model picks a more diverse set of features as most important



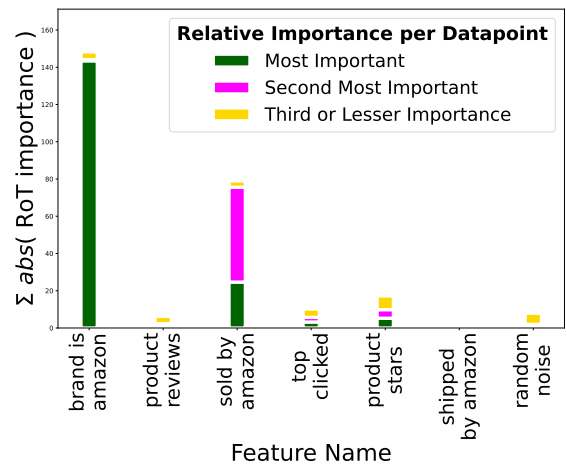
(c) The logistic regression model picks a different set of features as most important



(d) The L1 regularized logistic regression picks only “brand is amazon” as important

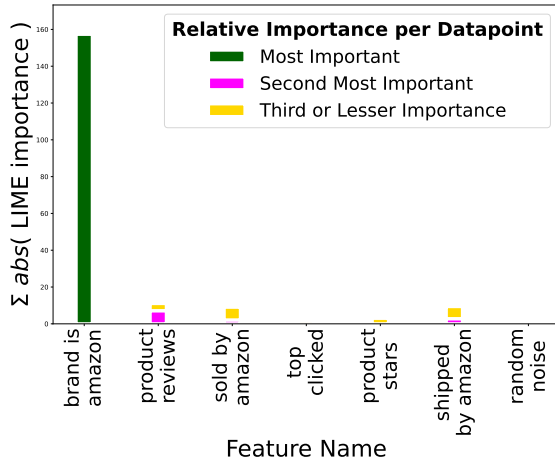


(e) The L2 regularized logistic regression model picks a different set of features as important

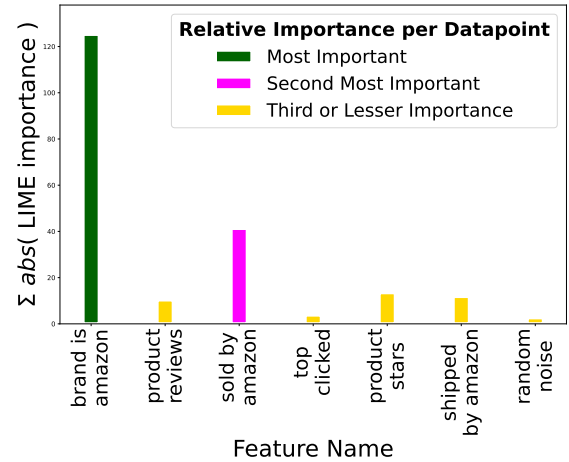


(f) RoT usually finds “brand is amazon” to be the most important, and often finds “sold by amazon” to be the second most important

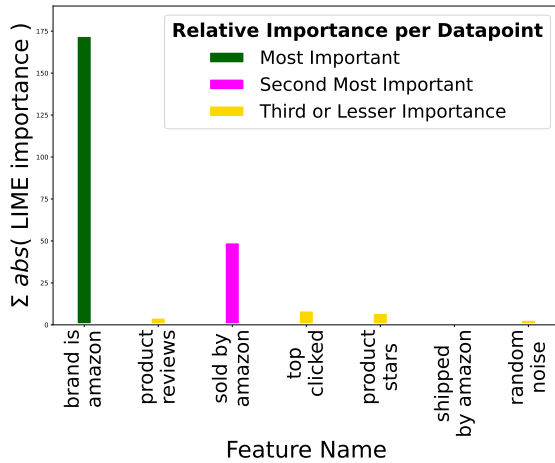
Figure 15. SHAP finds a different ordering of importances for the features, depending on the mimic model used, whereas RoT provides a consistent result and uncovers an additional insight highlighting the importance of “sold by amazon”



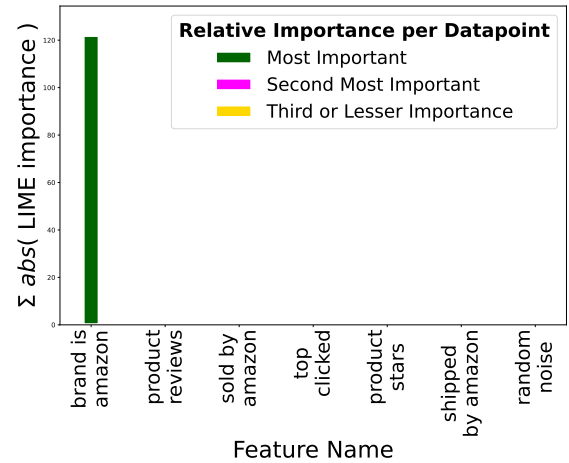
(a) The Markup model finds the most important feature to be “brand is amazon” with “sold by amazon” second



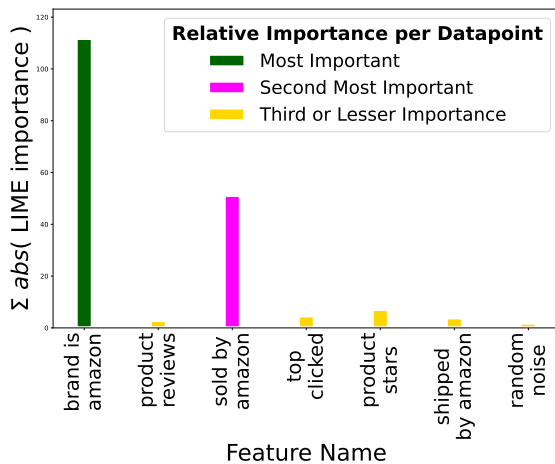
(b) The default random forest model picks a more diverse set of features as most important



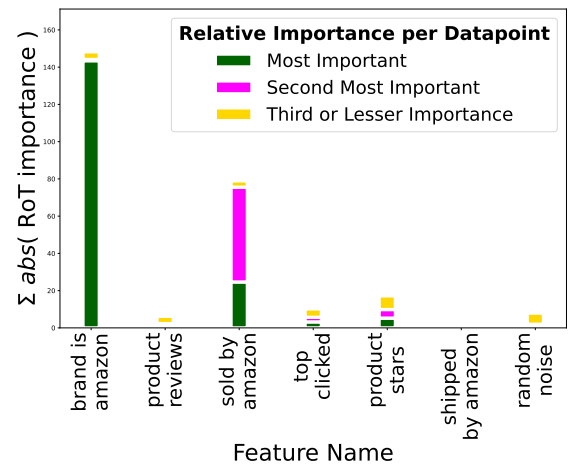
(c) The logistic regression model picks a different set of features as most important



(d) The L1 regularized logistic regression picks only “brand is amazon” as important

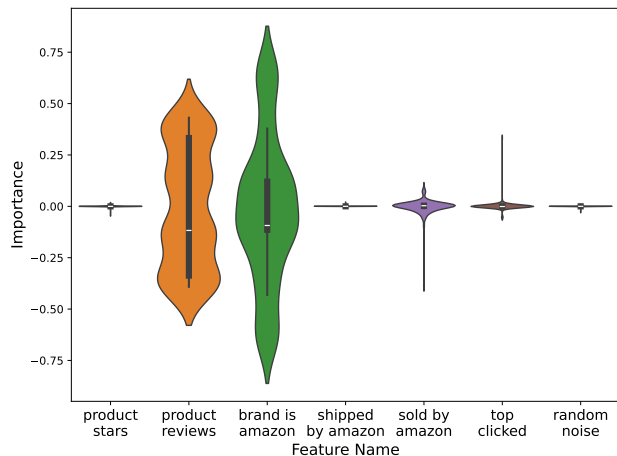


(e) The L2 regularized logistic regression model picks a different set of features as important

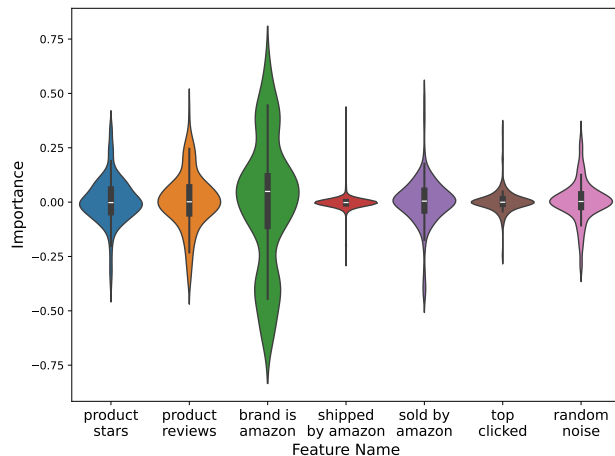


(f) RoT usually finds “brand is amazon” to be the most important, and often finds “sold by amazon” to be the second most important

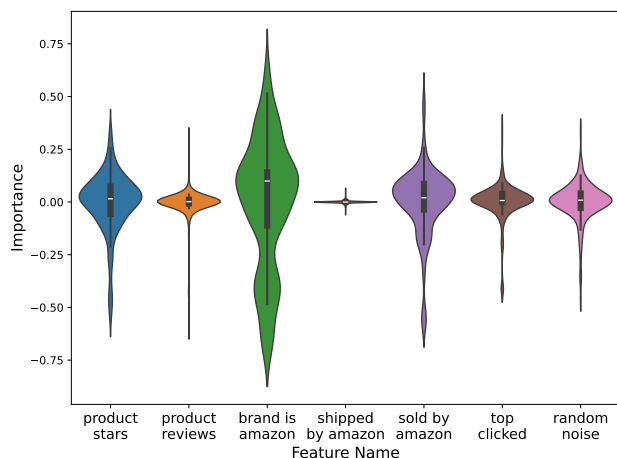
Figure 16. LIME finds a different ordering of importances for the features, depending on the mimic model used, whereas RoT provides a consistent result and uncovers an additional insight highlighting the importance of “sold by amazon”



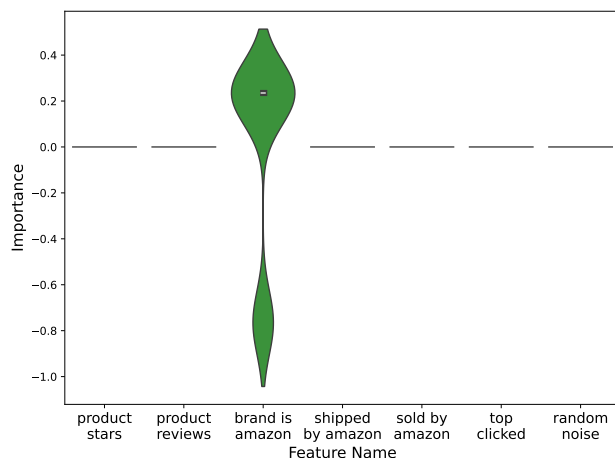
(a) The Markup model finds “brand is amazon” and “product reviews” to have importances of high magnitude



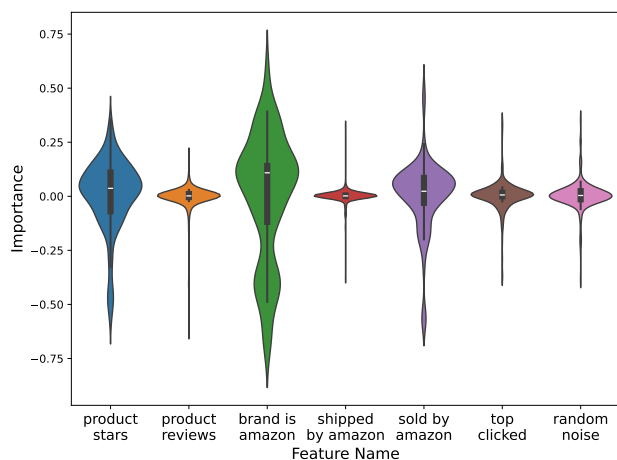
(b) The default random forest model generates a different profile of importance distributions



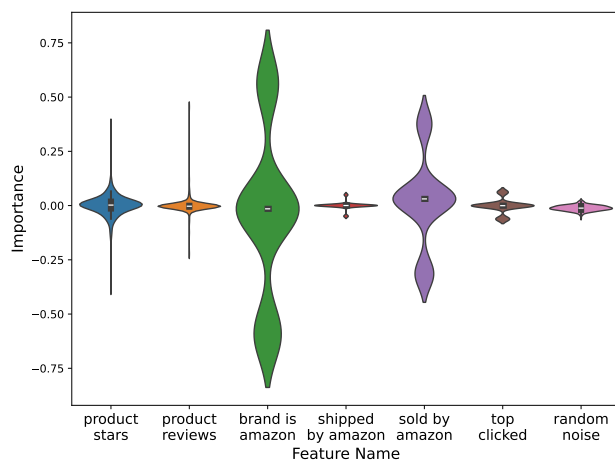
(c) The logistic regression model finds a different set of importances than both random forest models



(d) The L1 regularized regression model produces considers only “brand is amazon” to be important

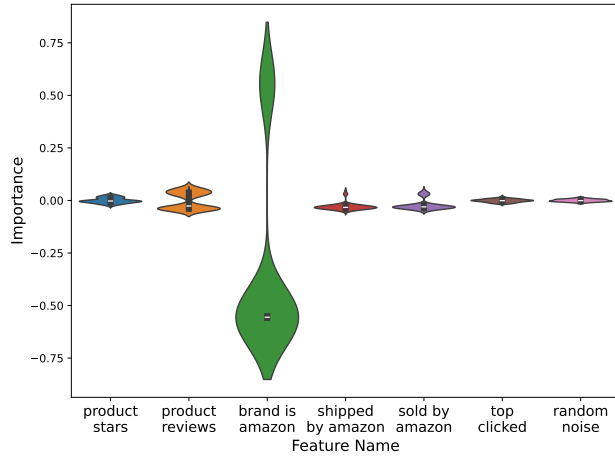


(e) The L2 regularized regression model produces a different importance distribution

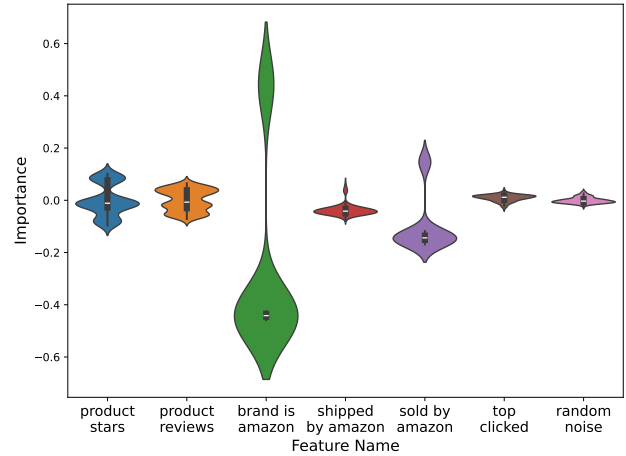


(f) RoT importances can be computed without any mimic models

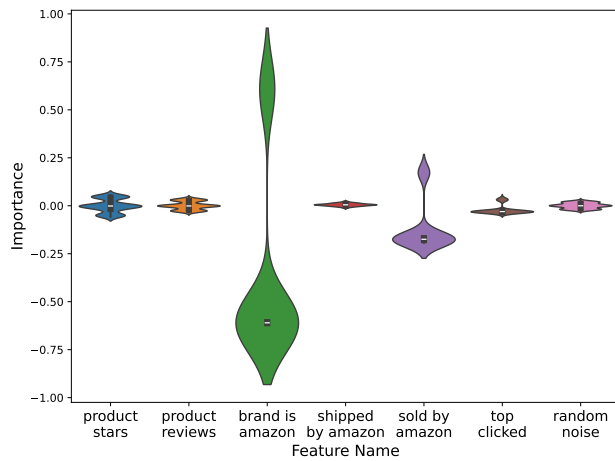
Figure 17. SHAP Importance Distributions for Input Features Can Vary by Choice of Mimic Model, whereas RoT provides a Single and Robust Set of Importance Distributions for all Features, and Cannot be Manipulated by Choice of Mimic Model



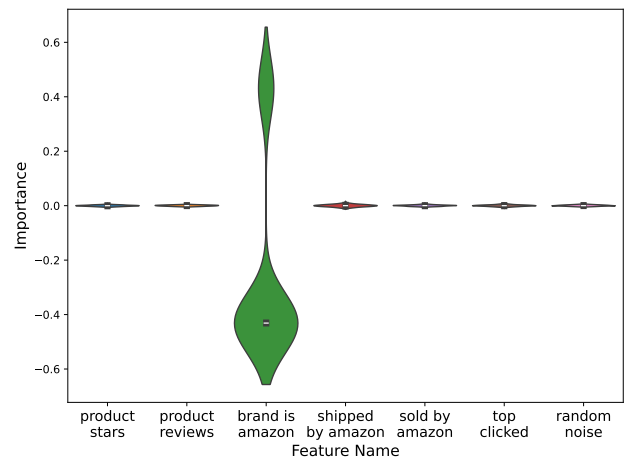
(a) The Markup model finds “brand is amazon” and “product reviews” to have importances of high magnitude



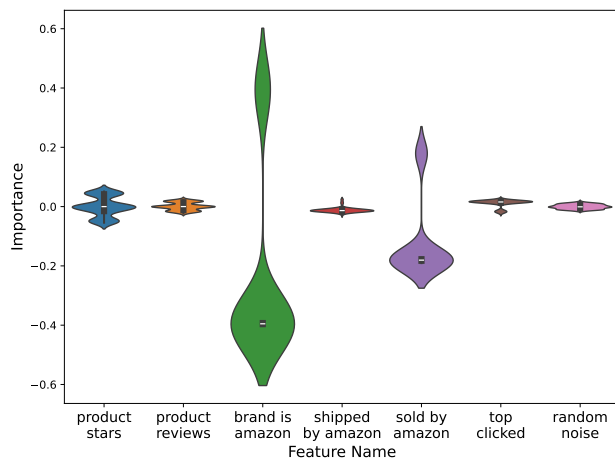
(b) The default random forest model generates a different profile of importance distributions



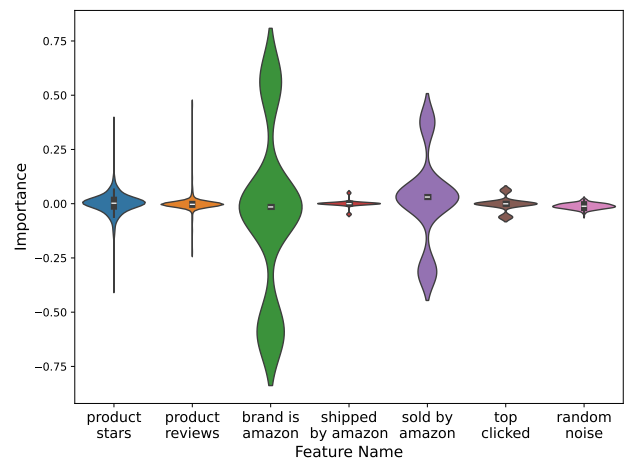
(c) The logistic regression model finds a different set of importances than both random forest models



(d) The L1 regularized regression model produces considers only “brand is amazon” to be important



(e) The L2 regularized regression model produces a different importance distribution



(f) RoT importances can be computed without a mimic model

Figure 18. LIME Importance Distributions for Input Features Can Vary by Choice of Mimic Model, whereas RoT provides a Single and Robust Set of Importance Distributions for all Features, and Cannot be Manipulated by Choice of Mimic Model

C.3 Scientific Discovery using XAI

Identifying Important Features in the presence of Misleading Features Table 7 effectively summarises the most important feature across datapoints, as picked by various explainers. This is the underlying explanation data that was plotted in the ternary distribution plots in Figure 6. To provide a more granular analysis, we also provide additional confusion matrices comparing the first, second, and third most important features as selected by various explainers across various datasets. in Tables 8, 9, 10, 11, and 12.

Dataset	Adversarial Model	Explainer	Percentage of Dataset where the Most Important Input Feature is:			
			a) Sensitive	b) Foil #1	c) Foil #2	d) Other
German Credit	m_L	LIME	0.0	84.0	na	16.0
	(1 foil)	RoT	100.0	0.0	na	0.0
	m_S	SHAP	0.0	100.0	na	0.0
	(1 foil)	RoT	100.0	0.0	na	0.0
COMPAS	m_L	LIME	0.0	100.0	na	0.0
	(1 foil)	RoT	100.0	0.0	na	0.0
	m_S	SHAP	1.9	90.9	na	7.1
	(1 foil)	RoT	100.0	0.0	na	0.0
	m_L	LIME	0.3	43.9	47.9	7.9
	(2 foils)	RoT	100.0	0.0	0.0	0.0
	m_S	SHAP	35.0	18.9	22.3	23.8
	(2 foils)	RoT	100.0	0.0	0.0	0.0
Communities and Crime	m_L	LIME	0.0	99.5	na	0.5
	(1 foil)	RoT	89.5	0.0	na	10.5
	m_S	SHAP	4.0	87.0	na	9.0
	(1 foil)	RoT	90.5	0.0	na	9.5
	m_L	LIME	0.0	14.5	74.5	11.0
	(2 foils)	RoT	91.0	0.0	0.0	9.0
	m_S	SHAP	1.0	49.0	50.0	0.0
	(2 foils)	RoT	90.0	0.0	0.0	10.0

Table 7. RoT Successfully Determines the Sensitive Feature as Most Important, Despite the Presence of Misleading Foil Features: adversarially crafted models m_L and m_S are designed to mislead LIME and SHAP respectively into selecting the Foil features as most important. For both adversarial models, we verify that RoT successfully identifies the Sensitive feature as most important. In each case, we present the percentage of predictions that yield a particular input feature as most important. **Successes are in bold green**, **adversarial failures in bold red**, and **non-adversarial failures in bold blue**.

Importance Rank	1	2	3
Synthetic Foil #1	90.9	7.3	1.8
race	1.9	55	19.9
other features	7.1	37.7	78.3

(a) SHAP explanations for adversarially crafted model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
race	100.0	0.0	0.0
other features	0.0	100.0	100

(b) RoT explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	100.0	0.0	0.0
race	0.0	23.1	18.8
other features	0.0	76.9	81.2

(c) LIME explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
race	100.0	0.0	0.0
other features	0.0	100.0	100.0

(d) RoT explanations for adversarial model

Table 8. COMPAS dataset, with a synthetic foil feature: Confusion matrices showing what percentage of explanations select particular features as the most, second most, and third most important features in explanations. Ideally, the most important feature recovered by an explainer should be the only one used (highlighted green), whereas the adversary tries to have the explainers always pick a different feature (highlighted red). The most important feature (first column) from each matrix is replicated in table 7

Importance Rank	1	2	3
Synthetic Foil #1	18.9	23.6	26.4
Synthetic Foil #2	22.3	27.0	18.4
race	35.0	23.6	12.6
other features	23.8	25.7	42.6

(a) SHAP explanations for adversarially crafted model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	1.6	8.1
Synthetic Foil #2	0.0	0.0	0.0
race	100.0	0.0	0.0
other features	0.0	98.4	91.9

(b) RoT explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	43.9	39.0	7.0
Synthetic Foil #2	47.9	27.5	10.7
race	0.3	3.1	13.3
other features	7.9	30.4	69.1

(c) LIME explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
Synthetic Foil #2	0.0	0.0	0.0
race	100.0	0.0	0.0
other features	0.0	100.0	100.0

(d) RoT explanations for adversarial model

Table 9. COMPAS dataset, with two synthetic foil features: Confusion matrices showing what percentage of explanations select particular features as the most, second most, and third most important features in explanations. Ideally, the most important feature recovered by an explainer should be the only one used (highlighted green), whereas the adversary tries to have the explainers always pick a different feature (highlighted red). The most important feature (first column) from each matrix is replicated in table 7

Importance Rank	1	2	3
Synthetic Foil #1	87.0	12.5	0.5
race	4.0	40.5	9.5
other features	9.0	42.5	90.0

(a) SHAP explanations for adversarially crafted model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
race	90.5	4.5	0.5
other features	9.5	95.5	99.5

(b) RoT explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	99.5	0.0	0.0
race	0.0	0.5	0.0
other features	0.5	99.5	100

(c) LIME explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
race	89.5	5.5	1.0
other features	10.5	94.5	99.0

(d) RoT explanations for adversarial model

Table 10. Communities and Crime dataset, with a synthetic foil feature: Confusion matrices showing what percentage of explanations select particular features as the most, second most, and third most important features in explanations. Ideally, the most important feature recovered by an explainer should be the only one used (highlighted green), whereas the adversary tries to have the explainers always pick a different feature (highlighted red). The most important feature (first column) from each matrix is replicated in table 7

Importance Rank	1	2	3
Synthetic Foil #1	49.0	23.5	7.5
Synthetic Foil #2	50.0	21.5	14.5
race	1.0	21.5	15.5
other features	0.0	33.5	62.5

(a) SHAP explanations for adversarially crafted model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
Synthetic Foil #2	0.0	0.0	0.0
race	90.0	4.5	1.5
other features	10.0	95.5	98.5

(b) RoT explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	14.5	23.5	8.5
Synthetic Foil #2	74.5	11.5	3.5
race	0.0	0.5	1.0
other features	11.0	64.5	87.0

(c) LIME explanations for adversarial model

Importance Rank	1	2	3
Synthetic Foil #1	0.0	0.0	0.0
Synthetic Foil #2	0.0	0.0	0.0
race	91.0	4.0	1.0
other features	9.0	96.0	99.0

(d) RoT explanations for adversarial model

Table 11. Communities and Crime dataset, with two synthetic foil features: Confusion matrices showing what percentage of explanations select particular features as the most, second most, and third most important features in explanations. Ideally, the most important feature recovered by an explainer should be the only one used (highlighted green), whereas the adversary tries to have the explainers always pick a different feature (highlighted red). The most important feature (first column) from each matrix is replicated in table 7

Importance Rank	1	2	3
LoanRate % of Income	100.0	0.0	0.0
gender	0.0	40.0	5.0
other features	0.0	60.0	95.0

(a) SHAP explanations for adversarially crafted model

Importance Rank	1	2	3
LoanRate % of Income	0.0	0.0	0.0
gender	100.0	0.0	0.0
other features	0.0	100.0	100.0

(b) RoT explanations for adversarial model

Importance Rank	1	2	3
LoanRate % of Income	84.0	0.0	0.0
gender	0.0	1.0	1.0
other features	16.0	99.0	99.0

(c) LIME explanations for adversarial model

Importance Rank	1	2	3
LoanRate % of Income	0.0	0.0	2.0
gender	100.0	0.0	0.0
other features	0.0	100.0	98.0

(d) RoT explanations for adversarial model

Table 12. German credit dataset, with an engineered foil feature (LoanRate as % of Income): Confusion matrices showing what percentage of explanations select particular features as the most, second most, and third most important features in explanations. Ideally, the most important feature recovered by an explainer should be the only one used (highlighted green), whereas the adversary tries to have the explainers always pick a different feature (highlighted red). The most important feature (first column) from each matrix is replicated in table 7

D Runtimes and Computational Efficiency

Tables 13 and 14 document RoT runtimes against SHAP and LIME across all our experiment with tabular data. These are visualised in Figure 8.

Data & Model Details (+ Num. Explanations)			RoT Time init (s)	SHAP Time init (s)	RoT Time total (s)	SHAP Time total (s)	Speedup Factor
AI Auditing	Markup RF	283	0.6358	0.0690	0.6359	295.41	464.54
Uninformative Features	Pima Diabetes FairPCA LR	154	1.0191	0.0007	1.0191	9.63	9.45
Adversarial Attack	German Credit	100	0.4273	0.0017	0.4274	32.12	75.17
	CC (1 foil)	200	0.7037	0.0017	0.7038	138.99	197.47
	CC (2 foils)	200	0.6782	0.0022	0.6783	136.19	200.78
	COMPAS (1 foil)	618	0.6126	0.0021	0.6127	69.86	114.03
	COMPAS (2 foils)	618	0.6091	0.0057	0.6092	68.43	112.32
OpenXAI Benchmark	German Credit LR	800	0.2621	0.0013	0.2768	233.55	843.88
	German credit ANN	800	0.2596	0.0009	0.2741	169.50	618.49
	COMPAS LR	4937	0.3344	0.0007	0.4127	189.93	460.23
	COMPAS ANN	4937	0.3243	0.0011	0.4049	217.48	537.14
	HELOC LR	7896	0.6619	0.0019	0.7843	522.34	666.00
	HELOC ANN	7896	0.6473	0.0010	0.7733	413.89	535.20
	Adult Income LR	36177	2.1747	0.0015	2.7339	1437.98	525.98
	Adult Income ANN	36177	2.0596	0.0006	2.6237	1668.77	636.04
Final Average	-	6786.2	0.7606	0.0061	0.8647	373.60	432.08

Table 13. Comparison of SHAP and RoT run-times across experiments. For each experiment we denote the dataset name and model type (Logistic Regression or Artificial Neural Network). The per-explanation run-time of RoT is negligible, making it much faster than SHAP despite slower initialization times. Consequently, RoT provides a greater speedup for larger datasets with more explanations to compute.

Data & Model Details (+ Num. Explanations)			RoT Time init (s)	LIME Time init (s)	RoT Time total (s)	LIME Time total (s)	Speedup Factor
AI Auditing	Markup	283	0.6358	0.0025	0.6359	21.95	34.52
Uninformative Features	Pima Diabetes FairPCA LR	154	1.0191	0.0020	1.0191	1.40	1.37
Adversarial Attack	German Credit	100	0.6934	0.0012	0.6935	2.39	3.45
	CC (1 foil)	200	1.0922	0.0008	1.0923	86.02	78.75
	CC (2 foils)	200	0.9712	0.0008	0.9713	84.87	87.37
	COMPAS (1 foil)	618	0.9836	0.0025	0.9838	19.95	20.28
	COMPAS (2 foils)	618	0.6865	0.0025	0.6866	20.87	30.40
OpenXAI Benchmark	German Credit LR	800	0.2621	0.0003	0.2768	1071.57	3871.97
	German Credit ANN	800	0.2596	0.0004	0.2741	1073.84	3918.41
	COMPAS LR	4937	0.3344	0.0004	0.4127	246.86	598.18
	COMPAS ANN	4937	0.3243	0.0017	0.4049	420.53	1038.64
	HELOC LR	7896	0.6619	0.0006	0.7843	2309.85	2945.13
	HELOC ANN	7896	0.6473	0.0063	0.7733	3991.78	5161.80
	Adult Income LR	36177	2.1747	0.0012	2.7339	2057.12	752.44
	Adult Income ANN	36177	2.0596	0.0015	2.6237	7396.13	2818.98
Final Average	-	6786.2	0.8537	0.0016	0.9577	1253.67	1308.99

Table 14. Comparison of LIME and RoT run-times across experiments. For each experiment we denote the dataset name and model type (Logistic Regression or Artificial Neural Network). The OpenXAI LIME implementation differs from the canonical by reducing background samples used per explanation from 5000 to 1000. In order to make up for this inconsistency we report extrapolated OpenXAI LIME runtimes here by multiplying with 5. Like with SHAP, RoT provides a greater speedup for larger datasets with more explanations to compute.

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